UNIVERSITY OF MESSINA

Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences (MIFT)

Bachelor's degree in Computer Science (L-31) · Data Analysis curriculum



REPORT ON INTERNSHIP ACTIVITIES

RobustCluster-Aβ

Reproducible analysis of amyloid-β40 conformational descriptors

Internship project: Clustering Framework for Intrinsically Disordered Proteins

STUDENT	Ali Erkan Ocaklı (student ID 555012)
HOST LABORATORY	Laboratory of Experimental and Computational Physics of Biosystems, MIFT
HOST SUPERVISOR	Prof. Valeria Conti Nibali
CO-SUPERVISOR	Andrea Ciccolo
ACADEMIC TUTOR	Prof. David Barilla
WORKLOAD	75 hours, completed

ACADEMIC YEAR 2025–2026

Abstract

This report describes a 75-hour curricular internship, completed in full, in the Laboratory of Experimental and Computational Physics of Biosystems at the University of Messina, on the project *Clustering Framework for Intrinsically Disordered Proteins*. After a preparatory study of protein structure, intrinsic disorder and molecular dynamics, with practice in GROMACS and VMD, the final task was to reproduce the clustering workflow of Cadenelli and colleagues for amyloid-β40, vary its parameters, and propose an independent analysis framework.

The supplied table contains 300,003 descriptor observations from three replicas of well-tempered metadynamics. An audit of the supplied script found problems in input handling, residual-cluster treatment, free-energy interpolation and weighted summaries, which a modular, tested reimplementation corrects. Run with the published parameters on all frames, the reimplementation finds 11 density-peak families that separate mainly by secondary structure and are shared unevenly between replicas. The partition is sensitive: changing only the initialization seed gives adjusted Rand indices (ARI) of 0.44–0.68 with the reference partition. Because the free-energy surface does not state its unit, populations were computed assuming $\mathrm{kJ\,mol^{-1}}$; reweighting then removes the helix-rich families and concentrates 77.7% of the weight on three compact or extended families, a pattern that persists if the file is read as $\mathrm{kcal\,mol^{-1}}$.

As the independent framework, Gaussian mixtures were fitted with and without solvent-accessible surface area (SASA), which correlates strongly with radius of gyration ($r = 0.848$). Across fifty converged fits, median agreement between the two feature sets was 0.864 at three components but 0.457 at six, so whether SASA is redundant depends on the model. Each 100,001-frame series yields only 3.2–28.7 effective samples; all results are therefore descriptive statements about a biased, correlated sample.

Keywords: curricular internship; amyloid-β40; intrinsically disordered peptides; well-tempered metadynamics; clustering; Gaussian mixtures; reproducibility.

Internship at a glance

THE INTERNSHIP

- **Host.** Laboratory of Experimental and Computational Physics of Biosystems, MIFT, University of Messina.
- **Supervision.** Prof. Valeria Conti Nibali (host supervisor), Andrea Ciccolo (co-supervisor), Prof. David Barilla (academic tutor).
- **Workload and period.** 75 hours completed; registered period 25 May–30 September 2026; training project signed by the degree coordinator on 25 June 2026.
- **Tools.** GROMACS, VMD, Python (NumPy, SciPy, pandas, scikit-learn, Matplotlib), pytest, Snakemake, Git/GitHub, LaTeX.

KEY FINDINGS

- **The supplied data pass every structural check.** 300,003 frames form exactly three replicas of 100,001 frames at 20 ps; the FES is a complete grid containing every descriptor query (Section 6.3 and Table 5).
- **The published workflow runs, but its partition is sensitive.** With the published parameters, all frames form 11 families; changing only the seed gives adjusted Rand indices (ARI) of 0.44–0.68 with that partition (Section 6.4).
- **Reweighting reshapes the families (assumed FES unit).** Under the kJ mol^{-1} assumption, three families carry 77.7% of the weight and the helix-rich families almost vanish; the pattern survives the kcal mol^{-1} alternative (Section 6.5).
- **Saved frames are far from independent.** Each 100,001-frame series yields only 3.2–28.7 effective samples, and all twelve estimates fail the reliability screen (Section 6.3).
- **Correlation does not make SASA redundant.** Dropping SASA preserves the three-component partition (ARI 0.862–0.868; 95.3% of frames keep matched labels), but at six components median agreement falls to 0.457 (Section 6.6).
- **A single fit can mislead.** At three components seed-to-seed ARI is 0.988–1.000; at two it drops as low as 0.011, which is why every setting was repeated with five seeds.

Contents

Abstract i

Internship at a glance ii

Abbreviations v

1 Introduction 1
- 1.1 Context . . . 1
- 1.2 Internship task . . . 1
- 1.3 Planned and actual work . . . 1
- 1.4 Structure of the report . . . 1

2 Host institution and internship framework 3
- 2.1 Host laboratory and supervision . . . 3
- 2.2 Organization and administrative procedure . . . 3
- 2.3 Programme and timeline . . . 3

3 Scientific and theoretical background 5
- 3.1 Amyloid-β40 and intrinsically disordered proteins . . . 5
- 3.2 Molecular dynamics and the sampling problem . . . 5
- 3.3 Well-tempered metadynamics and free-energy surfaces . . . 5
- 3.4 Statistical reweighting . . . 6
- 3.5 Descriptors and density-based clustering . . . 6

4 Tools, data and methods 7
- 4.1 Tools and technologies . . . 7
- 4.2 Supplied data . . . 7
- 4.3 Reproduction workflow . . . 7
- 4.4 FES interpolation and statistical weights . . . 8
- 4.5 Diagnostics for the independent analysis . . . 9
- 4.6 Reproducibility . . . 9

5 Activities carried out 10
- 5.1 Preparatory study . . . 10
- 5.2 Practical work with GROMACS and VMD . . . 10
- 5.3 Task 1: running and auditing the supplied script . . . 10
- 5.4 Task 2: varying the parameters . . . 11
- 5.5 Task 3: an independent analysis framework . . . 12
- 5.6 Software practice . . . 12

6 Results 13
- 6.1 What the available data support . . . 13
- 6.2 Shared variation in the descriptors . . . 13
- 6.3 Dependence between saved observations . . . 13
- 6.4 Density-peak families in the real data . . . 14
- 6.5 Weighted family populations under an assumed FES unit . . . 16
- 6.6 What changes when SASA is removed . . . 18

6.7 Interpretation 18

7 Skills acquired and critical reflection 19
7.1 Objectives and outcomes 19
7.2 Skills acquired 19
7.3 Related personal projects 19
7.4 Critical reflection 20

8 Conclusions and future work 21

Signatures 22

Bibliography 23

A Parameters and settings 25

B Mathematical derivations and estimator checks 27

C Supplementary diagnostics 28

D Code audit summary 32

E Reproducibility and software environment 34

F Personal projects and source-code evidence 35

Abbreviations

Term	Meaning	Term	Meaning
ACF	Autocorrelation function	HPC	High-performance computing
ARI	Adjusted Rand index	MCP	Model Context Protocol
BIC	Bayesian information criterion	MD	Molecular dynamics
CI	Continuous integration	MIFT	Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences
CV	Collective variable	PCA	Principal component analysis
DPC / DPA	Density-peak clustering / Density Peaks Advanced	SASA	Solvent-accessible surface area
ESS	Effective sample size	SD	Standard deviation
FES	Free-energy surface	SEM	Standard error of the mean
GMM	Gaussian mixture model	WT-MetaD	Well-tempered metadynamics

1 Introduction

This chapter states the internship task, how the work departed from the original proposal, and how the report is organized.

1.1 Context

Clustering molecular descriptors is a familiar data-analysis problem: it groups rows with similar measurements. Interpretation requires care, because consecutive observations may be correlated, simulation bias changes configuration frequencies, and descriptors can hide structural differences. An intrinsically disordered peptide such as amyloid-β40 (Aβ40) is described by many configurations rather than one fixed structure [1], and clustering is a natural way to summarize that diversity.

Cadenelli and colleagues studied monomeric Aβ40 and Aβ42 ensembles with well-tempered metadynamics (WT-MetaD), density-peak clustering (DPC), consensus between local clusters and statistical reweighting [2]; that study is called *the primary article* below. Two of the authors supervised this internship, and their clustering script, descriptor table and free-energy surface (FES) for Aβ40 became its working material.

1.2 Internship task

After the preparatory study described in Chapter 2, the supervisors assigned the final task on 18 September 2026 in three parts:

1. **Reproduce.** Run the supplied script and reproduce the published results.
2. **Vary.** Modify its parameters and observe the effects.
3. **Propose.** Independently propose and explore another clustering framework, or other analysis procedures learned during the Bachelor's programme. The supervisors described this as the most important part.

The task led to three working questions: can the analysis be reconstructed and tested; what do the descriptors reveal about redundancy and temporal dependence; and does removing the solvent-accessible surface area (SASA), the descriptor most correlated with compactness, preserve the fitted groups?

1.3 Planned and actual work

The published internship proposal described a broader project, in which all-atom simulations would be run with GROMACS and the effect of different solvents analysed. The work actually carried out used the supplied descriptor data, as the 18 September assignment specified; no new simulations or solvent comparison are reported. Within that scope, all three parts of the task were carried out: the published clustering workflow was run on the full data with the published parameters and its parameters were varied (Sections 5.3 and 5.4), and an independent framework was explored (Section 5.5).

1.4 Structure of the report

Chapter 2 describes the host laboratory, supervision and programme. Chapter 3 summarizes the scientific background needed to read a biased simulation, and Chapter 4 the tools, data and methods. Chapter 5 gives a detailed account of the activities carried out, block by block, and Chapter 6 reports

the completed analyses. Chapter 7 reflects on the skills acquired against the objectives, and Chapter 8 concludes.

Five limitations apply to the whole report; they are stated here once and referred to later as *Scope 1–5*.

SCOPE AND LIMITATIONS

1. **Sampling.** The descriptors come from WT-MetaD, so observed frequencies are biased and trajectory time is biased time. No equilibrium probabilities or kinetic rates are inferred from them.
2. **Weights.** The supplied FES is a complete, numerically valid grid, but its energy unit and replica-combination recipe are undocumented. Weighted populations are computed under an explicit assumption (kJ mol^{-1}, three replicas combined) and are marked as such (Section 6.5).
3. **Interpretation.** Clusters and Gaussian components are geometric groupings of four descriptors; none is identified as a metastable state or given biological meaning.
4. **Uncertainty.** Frames are strongly autocorrelated; reported minimum–maximum ranges describe the runs performed and are not confidence intervals.
5. **Evidence.** A passing test shows that code is correct on inputs with known answers; it is not an Aβ40 result.

2 Host institution and internship framework

This chapter describes where the internship took place, who supervised it and how the programme ran.

2.1 Host laboratory and supervision

The internship was hosted by the Laboratory of Experimental and Computational Physics of Biosystems of the Department of Mathematical and Computer Sciences, Physical Sciences and Earth Sciences (MIFT) at the University of Messina, on the project *Clustering Framework for Intrinsically Disordered Proteins*. Table 1 gives the administrative frame recorded in the signed training project.

Table 1 Administrative frame of the internship, as recorded in the signed training project and the published internship proposal.

Item	Entry
Student	Ali Erkan Ocaklı, student ID 555012; Computer Science (L-31), Data Analysis curriculum, enrolled 2023/2024, third year in 2025/2026
Host	University of Messina, Department MIFT; Laboratory of Experimental and Computational Physics of Biosystems
Host supervisor	Prof. Valeria Conti Nibali
Co-supervisor	Andrea Ciccolo, PhD student in Physics
Academic tutor	Prof. David Barilla
Workload	75 hours, completed; registered period 25 May–30 September 2026
Approval	Training project signed by the degree coordinator on 25 June 2026

Prof. Valeria Conti Nibali supervised the project and Andrea Ciccolo, a PhD student in Physics, followed the day-to-day work as co-supervisor. Both are co-authors of the Aβ40/Aβ42 study whose clustering script and data form the case study of this report [2], so the final assignment came directly from ongoing research rather than from a textbook exercise. The signed project states the aim as developing computational and analytical skills for the study of intrinsically disordered proteins, with a focus on the amyloid-β peptide.

2.2 Organization and administrative procedure

Study was done largely remotely: the co-supervisor circulated reading lists, lecture material and the analysis files by email, and regular meetings, held in person or online, were used to discuss the material. Before any activity could count towards the internship, the training project had to be signed by the student, the host supervisor and the academic tutor, approved by the degree coordinator and registered with the university protocol office. At the end, the internship regulation requires a report on the activities, a signed activity log and the assessment form of the host supervisor.

2.3 Programme and timeline

The programme moved from biochemistry to simulation and finally to an open analysis task. Table 2 lists the blocks as assigned in the supervisors' emails, and Figure 1 places them in time.

The clustering-project block defines the structure of the remaining chapters. Reproduction and parameter study correspond to Sections 5.3 and 5.4; the independent framework is described in Section 5.5, and its results in Chapter 6.

Table 2 Internship programme as assigned in the supervisors' emails. Dates mark when each block was assigned; the practical tool work is described in Section 5.2.

Block	Content
Proteins and disorder assigned 25 June 2026	Amino-acid structure and classification, ionization, peptide bonds; levels of protein structure, peptide-group conformation, folding and stability; intrinsically disordered proteins. Discussion meeting on 30 June.
Simulation assigned 1 July 2026	Force fields, molecular dynamics, protein dihedral angles and cis/trans conformations; introductory material on setting up and running an MD simulation. Practical work with GROMACS and VMD.
Clustering method resumed 4 September 2026	The supervisors' clustering framework, introduced before the summer break, studied through the Methods and Results sections on conformational clustering of the primary article, with background on density-peak and agglomerative clustering; online walkthrough of the Python implementation on 10 September.
Clustering project assigned 18 September 2026	(1) Run the supplied script and reproduce the published results; (2) vary its parameters and observe the effects; (3) independently propose and explore another clustering or analysis framework from the Bachelor's programme. Meetings on 16 and 24 September.
Report	A written report of about 20 pages (minimum 15) presenting the work.

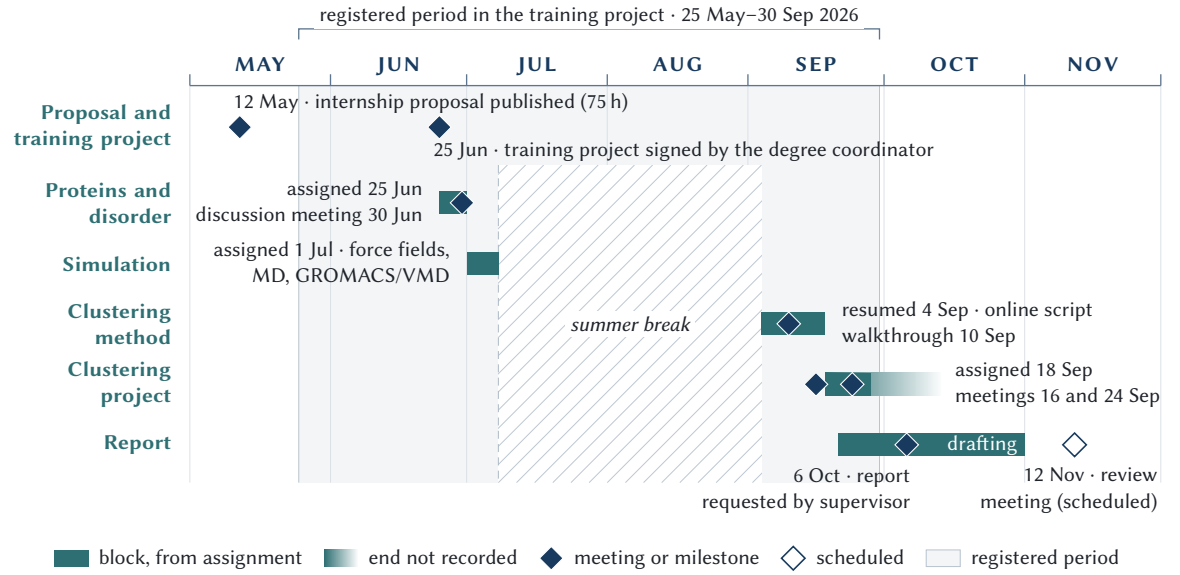


Figure 1 Internship timeline, May–November 2026. The grey band is the period registered in the training project. Teal bars run from each block's assignment to the next assignment, except that the simulation bar ends at the approximate start of the summer break and the report bar is an approximate drafting span; the faded bar marks an end date that was not recorded. Diamonds are meetings or milestones; the hollow diamond is the report review meeting scheduled for 12 November.

3 Scientific and theoretical background

This chapter summarizes the physics and statistics needed to read a biased simulation correctly.

3.1 Amyloid-β40 and intrinsically disordered proteins

Aβ40 is an intrinsically disordered peptide: no single structure describes it, and its state is an ensemble of conformations together with their statistical weights (Figure 2) [1, 2]. Observing a compact and an extended conformation therefore says nothing about how probable either is. Experiments usually report ensemble averages, and simulations can be combined with NMR or SAXS data when the observables are compatible [3].

3.2 Molecular dynamics and the sampling problem

Molecular dynamics integrates the equations of motion of all atoms, with forces from a force field; the primary article used GROMACS, the CHARMM36m force field and PLUMED [2, 4–6]. At equilibrium a configuration q has probability density proportional to $\exp[-\beta_T U(q)]$, with $\beta_T = 1/(k_B T)$ when the potential energy U is per molecule and $1/(RT)$ when it is per mole, so the energy unit must be known before any weight is computed. Saved frames are not independent observations: saving more often adds similar rows, not information, and treating dependent frames as independent makes an estimate look more precise than the sampling supports [7, 8].

3.3 Well-tempered metadynamics and free-energy surfaces

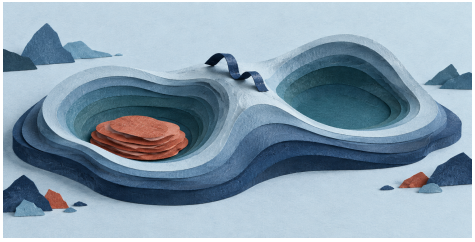
Metadynamics accelerates exploration by adding a history-dependent bias, built from Gaussian hills (Appendix B), along a few collective variables (CVs) $s(q)$, low-dimensional functions of the atomic coordinates chosen to follow slow structural changes. The free energy along the CVs is $F(s) = -\beta_T^{-1} \log p(s) + C$, where $p(s)$ is the equilibrium CV density; the map of F over the CVs is the free-energy surface (FES). In the well-tempered variant the bias grows ever more slowly, and the bias factor $\gamma = (T + \Delta T)/T$, with ΔT a chosen boost temperature, sets its long-time limit [9]:

$$V(s, t \to \infty) = -\left(1 - \frac{1}{\gamma}\right) F(s) + C. \tag{1}$$

In the primary article ($T = 310$ K, $\gamma = 10$) the bias offsets 90% of the free energy, so the sampled CV distribution, $p_b(s) \propto \exp[-\beta_T F(s)/\gamma]$, is the equilibrium distribution with every free-energy difference divided by ten. This is why the method explores efficiently, and why its raw frequencies must be reweighted before they can be read as populations; biased trajectory time does not give unbiased rates.



(a) A conformational ensemble



(b) Accumulating sampling bias

Figure 2 Conceptual illustrations (no data). (a) Flexible ribbons suggest conformational diversity; (b) layered deposits suggest the bias accumulated from Gaussian hills (Equation (7) of Appendix B).

Integrating out all other coordinates can merge different atomic arrangements at one point of this surface. PLUMED's `sum_hills` writes either the free energy or the negative bias, which differ by $\gamma/(\gamma-1) \approx 1.11$ for $\gamma = 10$ [10]: a grid can be numerically perfect and still be the wrong quantity. The offset-corrected bias `metad.rbias` is meaningful only together with the METAD settings that produced it [11].

3.4 Statistical reweighting

Weighting each sampled frame by $\exp(\beta_T V)$ undoes a quasi-static bias. Combined with Equation (1), this gives the final-bias weights used in the primary study:

$$\ell_i \equiv \log w_i = -\frac{1-1/\gamma}{RT} F(\alpha_i, \beta_i). \tag{2}$$

Here (α_i, β_i) are the two biased CVs of frame i (Section 4.4); the CV β is unrelated to β_T. At 310 K, $RT = 2.577\ \mathrm{kJ\,mol^{-1}}$, so with $\gamma = 10$ each $\mathrm{kJ\,mol^{-1}}$ of F lowers $\log w$ by 0.349; if the same numbers were $\mathrm{kcal\,mol^{-1}}$, every log weight would be 4.184 times larger. An additive constant in F cancels on normalization, but the unit does not, which is why it matters (Scope 2). Estimators based on the instantaneous bias, such as that of Tiwary and Parrinello, instead use $\ell_i = \mathrm{rbias}_i/(RT)$, with $\mathrm{rbias} = V(s,t) - c(t)$ [12, 13]; see also Bonomi, Barducci, and Parrinello [14].

3.5 Descriptors and density-based clustering

The four clustering descriptors summarize secondary structure ($n_\alpha = 1.1\alpha$ and $n_\beta = 2.5\beta$, rescaled biased CVs that approximate helical and β residue counts, not DSSP assignments [15]), compactness (radius of gyration R_g) and solvent exposure (solvent-accessible surface area, SASA). Because they differ in unit and spread, their scaling is part of the method (Chapter 4).

In density-peak clustering [16], a cluster centre combines a high local density ρ_i with a large distance δ_i to the nearest point of higher density. The published workflow searches for such centres in random subsets and keeps clusters that are large enough and have a sufficient mean silhouette, $(b-a)/\max(a,b)$, where a is a frame's mean distance to its own cluster and b to the nearest other one [2, 17]. Consensus then merges the local clusters into families, every frame is assigned to a family, and statistical weighting decides how much each family contributes.

4 Tools, data and methods

This chapter describes the tools, the supplied data and the methods used in the activities of Chapter 5.

4.1 Tools and technologies

Table 3 lists the software used during the internship and its role.

Table 3 Tools and technologies used during the internship.

Tool	Role in the internship	Where
GROMACS	Molecular-dynamics engine studied in the simulation block; also the engine used in the primary article [4]	Section 5.2
VMD	Visualization and structural measurement of molecular systems	Section 5.2
Python 3.12 with NumPy, SciPy, pandas, scikit-learn, Matplotlib	Data validation, clustering, diagnostics, Gaussian mixtures and figures [18–20]	Chapters 5 and 6
pytest and Ruff	Unit and end-to-end tests (38 passing) and static checks	Section 5.3
Snakemake	Workflow definition reusing the package functions [21]	Appendix E
Git, GitHub and GitHub Actions	Version control, public release and continuous integration (CI)	Section 5.3
LuaLaTeX, Biber	Typesetting of this report	—

4.2 Supplied data

The analysis starts from a supplied text table, not from atomic coordinates: 300,003 rows with time, an offset-corrected bias field and eight structural measurements. The script, the table and the FES are identified by SHA256 hashes. The published method is the scientific reference [2]; the supplied script provides implementation details, some of which differ from that description and are therefore checked.

A decrease in the time column marks a new replica. Validation requires exactly three segments with strictly increasing times, finite entries and complete coverage of the 1–3 µs interval (times read as picoseconds, as in the script and the article), giving 100,001 frames per replica at 20 ps; a malformed row rejects the whole file rather than being skipped. Every row keeps its file line, replica and frame index, so that partitions from different runs can be compared frame by frame. The first microsecond is not in the table, so its equilibration cannot be reassessed.

The clustering features are n_α, n_β, R_g (nm) and SASA (nm^2); the other columns are kept for characterization. They are scaled by pooled min–max normalization, $z_f = (x_f - a_f)/(b_f - a_f)$, where a_f and b_f are the minimum and maximum of feature f over all three replicas, so that all replicas and discovery groups share one coordinate system.

4.3 Reproduction workflow

Figure 3 summarizes the implemented reproduction. Frames are divided into random, non-overlapping groups of about 30,000, drawn from a recorded master seed [22]. Within a group,

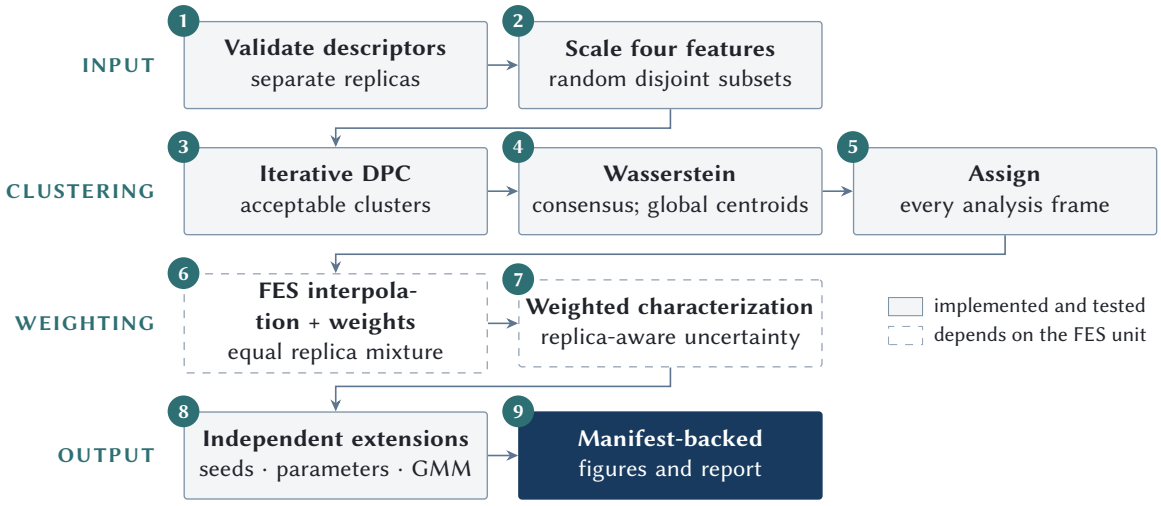


Figure 3 Implemented reproduction workflow. Accepted local clusters define global families, and every frame then receives a label. Dashed stages depend on the FES and were run under the assumed energy unit of Section 6.5.

using Euclidean distances in the scaled space, ρ_i is the number of frames within the cutoff d_c of frame i (itself included) and δ_i the distance to the nearest denser frame; equal densities are ordered by position in the shuffled group, and the densest frame receives the largest distance in the group. The k frames with the largest $\rho_i\delta_i$ become centres, and every other frame, taken in decreasing density, inherits the label of its nearest denser frame. Clusters are accepted only if they reach a minimum size and mean silhouette; the iterative protocol and its parameters are listed in Appendix A (Table 10). Distances are computed in row blocks into a full float32 matrix (about 3.6 GB for 30,000 frames), so memory still grows as N^2.

Consensus. Accepted local clusters are compared with the distance used in the supplied script,

$$D(A,B) = \left[\sum_{f=1}^{4} W_1(A_f, B_f)^2\right]^{1/2}, \tag{3}$$

where $W_1(A_f, B_f)$ is the one-dimensional first Wasserstein (earth mover's) distance between the unweighted values of descriptor f in clusters A and B [23]; D is not a joint four-dimensional transport distance. The baseline keeps the script's raw scales, in which R_g (pooled standard deviation 0.26 nm) counts about twenty times less than n_α and n_β (5.3 and 6.0, dimensionless) and SASA (4.8 nm^2); a normalized variant tests whether this matters. Average-linkage clustering of D is cut at the merge height where a smoothing spline ($s = 5$) fitted to the merge heights has its largest second derivative within the first 90% of merges, as in the supplied script.

Final assignment. Every frame, including discovery members and frames left unassigned, receives the label of the nearest family centroid (Euclidean, min–max space), each centroid being the unweighted mean of the family's member frames; there is no iterative refinement, so a frame can leave the family of its local cluster.

4.4 FES interpolation and statistical weights

The FES reader checks the grid structure, and the FES is interpolated bilinearly at each frame's CV values $\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$ with strict bounds [19, 24]; a query outside the grid is an error, not a

value to clip. Assuming the file is in kJ mol^{-1} (Scope 2), the final-bias log weights of Equation (2) are normalized within each replica before pooling,

$$\widetilde{w}_{ri} = \exp\left[\ell_{ri} - \mathrm{logsumexp}_j(\ell_{rj})\right], \qquad W_{ri} = \widetilde{w}_{ri}/3, \tag{4}$$

for frame i of replica r; the log-sum-exp avoids overflow, and each replica carries one third of the total weight. Weight concentration is summarized per replica by the Kish effective sample size (Equation (8)), and family populations are reported pooled and for each replica separately; the spread between the three replicas is the uncertainty statement used in Section 6.5.

4.5 Diagnostics for the independent analysis

Temporal correlation. For each replica and descriptor, the unweighted series is centred and its autocorrelation $\hat{\rho}(k)$ computed by FFT with $1/N$ normalization at every lag. Following Geyer's initial monotone sequence estimator [25], but pairing lags from 1 rather than 0, the pair sums $\Gamma_m = \hat{\rho}(2m-1) + \hat{\rho}(2m)$ are made non-increasing by a running minimum, $\tilde{\Gamma}_m = \min_{j \le m} \Gamma_j$, and summed up to M, the last pair before the first non-positive one:

$$g = \max\Big(1,\ 1 + 2\sum_{m=1}^{M} \tilde{\Gamma}_m\Big), \qquad N_{\mathrm{eff,corr}} = N/g, \qquad \tau_{\mathrm{int}} = g\Delta t/2. \tag{5}$$

Here g is the statistical inefficiency, $N_{\mathrm{eff,corr}}$ the effective sample size (ESS) and τ_{int} the integrated time, measured in biased simulation time; it exceeds the $(g-1)\Delta t/2$ of Chodera [7] by $\Delta t/2 = 10$ ps, negligible here. As project heuristics, an estimate is called reliable if a non-positive pair occurs, $N_{\mathrm{eff,corr}} \ge 50$, and the half-series means differ by at most 0.25 full-series standard deviations. Geyer's justification assumes a stationary, reversible chain, which a WT-MetaD trajectory is not, so the result serves as a screening diagnostic rather than a formal stationarity test [8].

Partition agreement. The adjusted Rand index (ARI) compares partitions independently of their label names [26]. It equals 1 for identical partitions, has expected value 0 for random labellings with the same cluster sizes, and can be negative. A one-to-one matching of families that maximizes total overlap (Hungarian algorithm) then supports Jaccard and centroid comparisons; unmatched families are retained. Seed variation measures algorithmic sensitivity, not uncertainty between independent replicas. The same comparison is used for the density-peak parameter study (Section 5.4) and for the Gaussian-mixture experiment (Section 5.5).

Feature redundancy. Pearson correlation and PCA of standardized descriptors describe the observed (unweighted, biased) feature geometry, not equilibrium relevance.

4.6 Reproducibility

Every command first writes a manifest with the configuration, input hashes, seeds and software versions, and records runtime and peak memory; the command-line interface and the Snakemake workflow call the same functions [21, 27]. Every data figure and table in this report can therefore be traced to one recorded calculation.

5 Activities carried out

This chapter gives a detailed account of the activities carried out, following the programme blocks of Table 2.

5.1 Preparatory study

The internship required more than applying a familiar statistical method to a new table. The assigned programme began with the concepts needed to interpret the data: amino acids, peptide bonds, protein structure and intrinsic disorder. Force fields, molecular dynamics and dihedral angles connected those concepts to simulation. The final task brought them together through examination of a clustering analysis and an independent investigation.

Two ideas from this block shaped all later work. Saved configurations need not be independent observations, and their frequency in a biased simulation differs from their equilibrium probability. Both connect the molecular setting to statistics and explain why the origin of the data matters as much as the clustering method.

5.2 Practical work with GROMACS and VMD

I used GROMACS, VMD and Python. Learning GROMACS was one of the more demanding parts of the internship for me, and one I came to enjoy as I became familiar with the software. The simulation block introduced the practical steps of setting up and running a molecular-dynamics simulation, and VMD was used to inspect and measure molecular structures. These exercises were training activities: the Aβ40 data analysed in this report were produced by the original study, not by simulations run during the internship. Working with both programs also led to two personal software projects that wrap them in a simpler interface (Section 7.3). Figure 16 in Appendix F places the tools in relation to each other and to these projects.

5.3 Task 1: running and auditing the supplied script

Running the supplied script

The audit separated three questions: what the paper describes, what the supplied code attempts, and what execution records show. The supplied script combines input reading, replica separation, clustering, weighting and plotting in one file, and an uncommented Italian sentence at line 437 stops it with a syntax error before anything runs. No original execution record was available, so the script is evidence of intent rather than of a completed run; it is preserved unchanged.

Identified methodological issues

The corrections in Table 4 concern which observations enter the analysis and how clusters are interpreted.

Random grouping was also corrected: the script draws its groups from a fixed list of 17 seeds, which runs out if more groups are requested, whereas the reimplementation derives independent streams from one recorded seed.

Verification and numerical tests

The 38 tests check each numerical operation on inputs with known answers, such as an analytic bilinear surface and weight normalization with unequal replica lengths, and run synthetic end-to-end

Table 4 Methodological corrections in the modular implementation. The original script remains unchanged.

Supplied-script behaviour	Correction and consequence
Takes the first three time-reset segments and skips some malformed rows.	Checks every row and requires exactly three segments, making incomplete or invalid input an explicit failure.
Sends rejected frames into consensus under a generic label.	Uses accepted memberships to discover global centres. Rejected frames receive final labels without being treated as a discovered family.
Permits missing FES cells; nearest-grid lookup can return a boundary value for an out-of-range query.	Requires a complete grid and bounded bilinear interpolation; queries outside the grid fail.
Weights relevance but uses unweighted descriptor means and standard deviations.	Weights cluster summaries consistently and normalizes each replica before pooling. Distribution width, replica SEM and bootstrap intervals remain distinct.

fixtures; two worked examples are given in Appendix D. On every push and pull request, a GitHub Actions workflow installs the package on Python 3.12 and runs the Ruff checks and the test suite; it uses no raw descriptors and no FES.

Remaining ambiguities

The paper does not fully specify the scaling used in consensus, how the replica-averaged FES was built, or the conventions behind every reported uncertainty, and an intermediate script version mentioned in the materials could not be located. Where the code was clearer, its choice was adopted and recorded as source-informed.

Running the corrected workflow on the real data

The three clustering steps (local density-peak discovery, Wasserstein consensus and final assignment) do not use statistical weights: only the final characterization of families needs the FES. A separate `cluster` command therefore runs these steps without reading the FES, and reports family sizes as raw frame fractions of the biased sample rather than populations. With the published parameters and group size, all 300,003 frames entered discovery in 10 random groups of about 30,000; 96 local clusters passed the size and silhouette criteria, and consensus merged them into 11 families (Section 6.4). The run took 14 minutes with a peak memory of 3.8 GiB. Rerunning with the same seed reproduces the partition exactly. This completes the run of the published clustering workflow; its families have not yet been compared with those reported in the primary article. The weighted populations were then computed under an explicitly stated assumption about the FES unit (Section 6.5).

5.4 Task 2: varying the parameters

With the workflow running on the real data, the parameter study could be carried out. Every variant was compared, on the same frame identities and using the adjusted Rand index, with the seed-42 partition of the same profile. Two kinds of variation were tested, each at the published full profile and at a reduced medium profile (three groups of 10,000 frames):

- *Initialization*: the random assignment of frames to discovery groups, with four further seeds derived from the master seed;

- *One parameter at a time*: the cutoff d_c scaled by 0.9 and 1.1, the silhouette threshold set to 0.25 and 0.35, and the minimum cluster size set to 200 and 400 frames.

A third comparison replaced the raw descriptor scales in the Wasserstein consensus with normalized ones, testing whether the under-weighted R_g changes the families. The results are reported in Section 6.4. Comparisons that still need further inputs were prepared but not run:

- *Time-dependent reweighting*: keep labels fixed and contrast final-bias weights with `metad.rbias` weights [12–14], once the metadynamics settings are known;
- *Alternative density clustering*: DPA [28]; ShapeGMM would additionally need structural coordinates [29];
- *External validation*: comparison with an experimentally constrained Aβ40 ensemble [3].

5.5 Task 3: an independent analysis framework

The supervisors asked for an alternative framework drawn from the Bachelor's programme. The data-mining course of the curriculum had covered correlation analysis, feature scaling and outlier detection; the independent framework builds on these and extends them with principal component analysis (PCA) and Gaussian mixture models (GMMs). The descriptor diagnostics suggested a concrete question for this framework: radius of gyration and SASA are strongly correlated (Chapter 6), so is SASA redundant for clustering? The experiment tests whether removing SASA changes assignments, comparing four descriptors with n_α, n_β and R_g using Gaussian mixtures:

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \Sigma_j), \qquad \sum_j \pi_j = 1. \tag{6}$$

In Equation (6), components approximate the sampled descriptor distribution; their mixture coefficients are not equilibrium weights. The scikit-learn implementation uses full covariance matrices [20].

The protocol was fixed before execution (Table 11): every fit uses the same 3,000 frames, 1,000 per replica; the scaling is estimated on those rows only and shared between the two feature sets; each of the five component counts $K = 2, \dots, 6$ is fitted with five seeds in both feature spaces, giving 50 fits; and all 300,003 frames are then labelled and compared with ARIs, five paired across feature sets and ten between seeds within each feature set.

Keeping the fitting sample fixed isolates the feature and seed effects.

5.6 Software practice

Small programming decisions also became explicit parts of the analysis. Feature scaling is fitted on a specified sample and reused for later observations. A test checks that a large evaluation value cannot alter that transformation. Original frame identities are retained so that label arrays refer to the same observations. These details determine whether a difference comes from the method under investigation or an unintended change in preprocessing.

Figure 15 and the scaling function in Appendix D show how the package fits together: inputs are validated once at the boundary, analysis functions receive explicit arrays and settings, and each command writes a manifest before producing tables or figures.

6 Results

This chapter reports the completed analyses: data and FES checks, descriptor relationships, temporal dependence, the density-peak families, their weighted populations and the feature experiment.

6.1 What the available data support

The calculations reported here use the supplied Aβ40 descriptors. Tests with generated data are discussed with the software validation and are not included as molecular observations. Table 5 distinguishes the completed analyses from the unresolved reproduction steps.

Table 5 Scope of the recorded calculations.

Analysis	Status and scope
Descriptor diagnostics	Complete; unweighted observations from three replicas
FES grid and query checks	Complete; energy unit and construction undocumented
GMM feature comparison	Complete; 50 fits on a fixed fitting sample
Paper DPC clustering	Complete; unweighted, published parameters and full profile
DPC parameter study	Complete; seeds and one-parameter changes
Weighted family populations	Complete under an assumed kJ mol^{-1} unit (Section 6.5)
Structural representatives	BLOCKED: needs coordinates and a frame map

Input validation found three segments of 100,001 rows. Times increase at the source-labelled spacing of 20 ps, and the data satisfy the column, finite-value and interval checks. The descriptor calculation, run `20260921T011959-583b4421`, took 2.28 s with a sampled peak worker-process memory of 418.2 MiB.

The complete FES file contains 2,801 by 1,301 grid points, giving 3,644,101 rows with five finite columns. Every expected coordinate pair is present and none is duplicated. The nonperiodic axes cover 0–28 and 0–13 at spacing 0.01, and all 300,003 converted queries fall inside the grid and yield finite bilinear interpolations. Figure 4 shows why this matters: the sampled frames occupy a triangular region of low free energy near small α and moderate β, while 61% of the grid sits on a flat maximum that lies largely outside the sampled region. The file is numerically usable; only its unit and construction stand between it and equilibrium weights (Scope 2).

6.2 Shared variation in the descriptors

The pooled R_g–SASA correlation is 0.848, against −0.456 between the two secondary-structure descriptors, and the relationship holds in every replica (per-replica r between 0.805 and 0.863). It is not a straight line: compact frames ($R_g < 1.3$ nm) form a dense core in which SASA rises steeply with R_g, while extended frames spread along a flatter band (Figure 9 in Appendix C). After standardization the first two principal components explain 51.8% and 36.3% of the variance, together 88.1% (Figure 10); the per-replica distributions of each descriptor are in Figure 11. This strong association is a reason to test whether SASA is needed for clustering (Section 6.6), not a reason to drop it in advance.

6.3 Dependence between saved observations

Each replica stays near a compact radius of gyration of about 1.0–1.2 nm for long stretches and leaves it only in a few extended excursions lasting tens to hundreds of nanoseconds of biased time (Figure 12 in Appendix C), so neighbouring frames, 20 ps apart, are nearly copies of each other. For the 12 replica–descriptor series, statistical inefficiencies range from 3,490 to 31,742 frames, giving

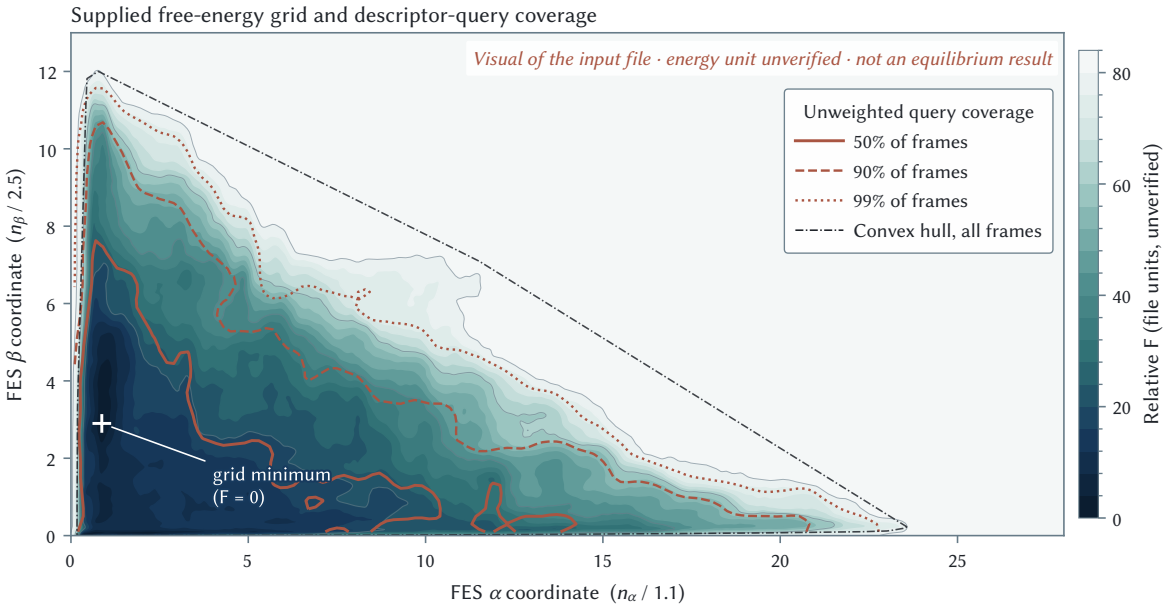


Figure 4 The supplied free-energy grid (minimum 0), plotted at every fourth grid point along each axis. Values stay in undeclared file units. Terracotta contours enclose 50%, 90% and 99% of the 300,003 unweighted descriptor queries ($\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$); the dash-dot line is their convex hull, and no query falls outside the grid.

correlation ESS values of only 3.2–28.7 per 100,001 frames (Table 12; autocorrelation functions in Figure 13). Every series falls below the required $N_{\text{eff,corr}} \geq 50$, and 11 of 12 shift by more than 0.25 standard deviations between their halves; n_α in replica 3 has the longest integrated time, 317 ns of biased time. Independent-frame error bars would therefore be unjustified. These diagnostics describe the supplied 1–3 µs data, not the convergence of the published runs, whose first microsecond and bias history are not available.

6.4 Density-peak families in the real data

Run with the published parameters, the workflow divides the 300,003 frames into 11 families (Figure 5 and Table 6). The largest holds 15.9% of the frames and the smallest 1.6%. The families separate mainly by secondary structure rather than by compactness. Nine of the eleven have mean radii of gyration between 1.04 and 1.40 nm and lie close together in Figure 5A, whereas in the (n_α, n_β) plane of Figure 5B their centres spread from helix-rich ($\bar{n}_\alpha$ up to 16.6) to β-rich ($\bar{n}_\beta$ up to 19.3) regions. Only two families are extended, with mean R_g of 1.58 and 1.81 nm; together they hold 11.5% of the frames.

The replicas do not share the families evenly. Family 3, the most helix-rich, holds 18.8% and 19.5% of the frames of replicas 1 and 3 but only 0.3% of replica 2, while the β-rich family 5 holds 22.9% of replica 2 and 2.9% of replica 3; the largest difference between replicas, for family 5, is 19.9 percentage points. This agrees with the temporal diagnostics of Section 6.3: each replica spends long stretches in a few regions, so the three replicas sample the landscape in different proportions.

The partition is sensitive to how it is computed (Figure 6). Changing only the initialization seed gives ARIs of 0.44–0.68 with the seed-42 partition and between 12 and 14 families. Changing one parameter at a time gives ARIs of 0.57–0.74 and between 11 and 14 families. The amount of data entering discovery matters as well: the medium profile, which uses 30,000 frames for discovery instead of all of them, finds only 3 families. Replacing the raw descriptor scales in the consensus with normalized ones left the medium-profile partition unchanged (ARI 1.00), so the low raw-scale weight of R_g does not affect the result there.

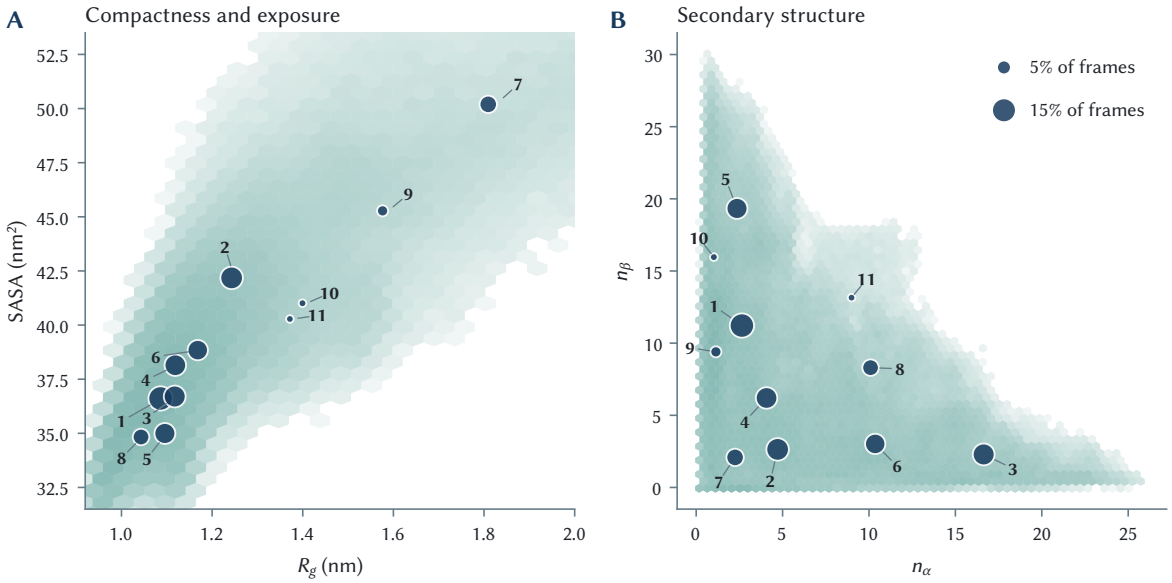


Figure 5 Density-peak families of the published workflow (published parameters, full profile, seed 42) on the unweighted descriptor landscape. Bubbles mark family means, numbered as in Table 6; their area is proportional to the raw frame fraction. The background shows the density of all frames; panel A is zoomed on the region that contains the family means.

Table 6 Density-peak families: fractions of frames, overall and by replica, and unweighted descriptor means.

	Fraction of frames (%)				Unweighted mean			
Family	All	Rep. 1	Rep. 2	Rep. 3	$\bar{n}_\alpha$	$\bar{n}_\beta$	$\bar{R}_g$ (nm)	$\overline{\mathrm{SASA}}$ (nm²)
1	15.9	10.4	26.1	11.3	2.6	11.2	1.09	36.6
2	13.7	14.0	11.2	16.0	4.7	2.6	1.24	42.2
3	12.9	18.8	0.3	19.5	16.6	2.3	1.12	36.7
4	12.5	8.7	14.6	14.2	4.1	6.2	1.12	38.1
5	11.8	9.6	22.9	2.9	2.4	19.3	1.10	35.0
6	11.1	19.1	4.8	9.5	10.4	3.0	1.17	38.8
7	8.2	4.2	9.3	11.0	2.3	2.1	1.81	50.2
8	7.4	11.6	2.1	8.6	10.1	8.3	1.04	34.8
9	3.3	1.0	4.5	4.5	1.1	9.4	1.58	45.3
10	1.6	1.0	1.3	2.4	1.0	16.0	1.40	41.0
11	1.6	1.7	3.0	0.0	9.0	13.2	1.37	40.3

Fractions count frames of the biased sample; they are not equilibrium populations (Scope 1–2). Families are numbered by decreasing frame fraction; the run's internal labels, in the same order, are 9, 10, 5, 11, 1, 7, 2, 8, 3, 4, 6.

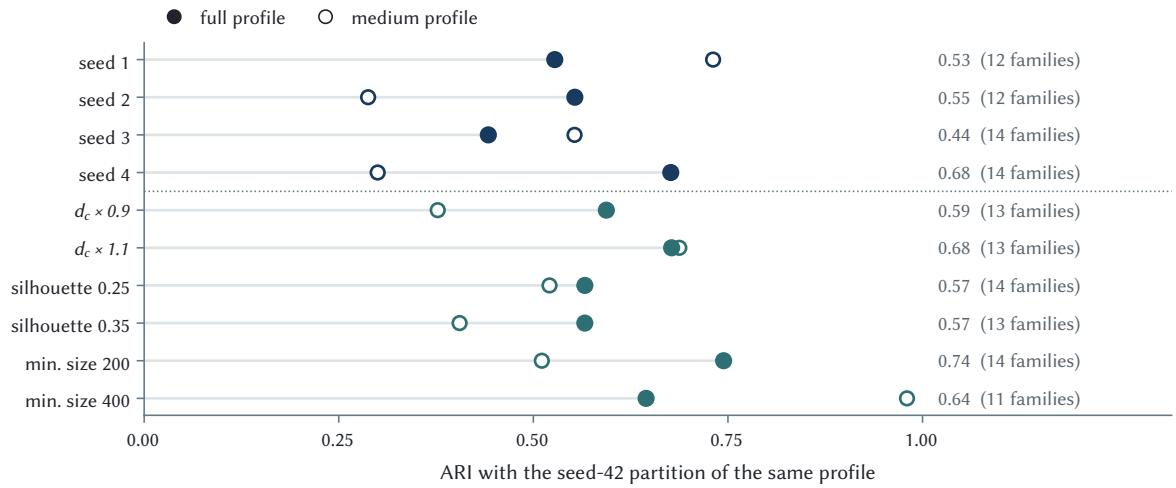


Figure 6 Sensitivity of the density-peak partition. Each row changes either the initialization seed (navy) or one parameter (teal) of the published configuration. Filled dots: ARI with the full-profile seed-42 partition (11 families); open dots: ARI with the medium-profile seed-42 partition (3 families). Labels give the full-profile ARI and family count.

6.5 Weighted family populations under an assumed FES unit

> **ASSUMPTION**
>
> The supplied FES does not state its energy unit or how the three replicas were combined. The populations in this section assume that `FES_AB_40.dat` is the `sum_hills` free energy (`file.free`) of the three replicas combined, in kJ mol^{-1}. This is the PLUMED default, the unit used by the supplied script and the unit of the primary article, but it has not been confirmed by the authors. If the assumption is wrong, these populations are wrong; the kcal mol^{-1} column of Table 7 shows how much they would change.

Under this assumption, the final-bias weights of Equations (2) and (4) were applied to the same 11 families; rerunning the clustering inside the weighted workflow reproduced the unweighted labels exactly. Reweighting changes the picture substantially (Figure 7 and Table 7). Family 1, compact with little helix and moderate β content, carries 47.4% of the equilibrium weight, and the three heaviest families (1, 4 and 7) together 77.7%. The helix-rich families 3, 6, 8 and 11 and the strongly β-rich family 5, which together hold 45% of the frames, keep less than 1% of the weight (at most 1.7% in any replica): the bias drove the simulations into these regions, but under this reweighting they carry almost no equilibrium weight. The two extended families gain weight, from 11.5% of the frames to 23.9%. The largest change for a single family is 31.5 percentage points.

The weights are strongly concentrated. The Kish effective sample size is 3,277–9,551 per replica of 100,001 frames; it measures weight concentration only and ignores the temporal correlation of Section 6.3, so the independent information is far smaller. The replicas also disagree: the population of family 1 ranges from 39.8% to 61.8%. If the file were in kcal mol^{-1} (Equation (2)), populations would shift by up to 14.6 percentage points: family 1 would still dominate, with 62.0%, and the helix-rich and strongly β-rich families would still vanish, but family 7 would fall to 2.4% and family 2 to 0.4%, and the extended families would hold 13.1%. The dominance of family 1 and the loss of those families therefore do not depend on which of the two units is correct; the gain of the extended families and the individual percentages do.

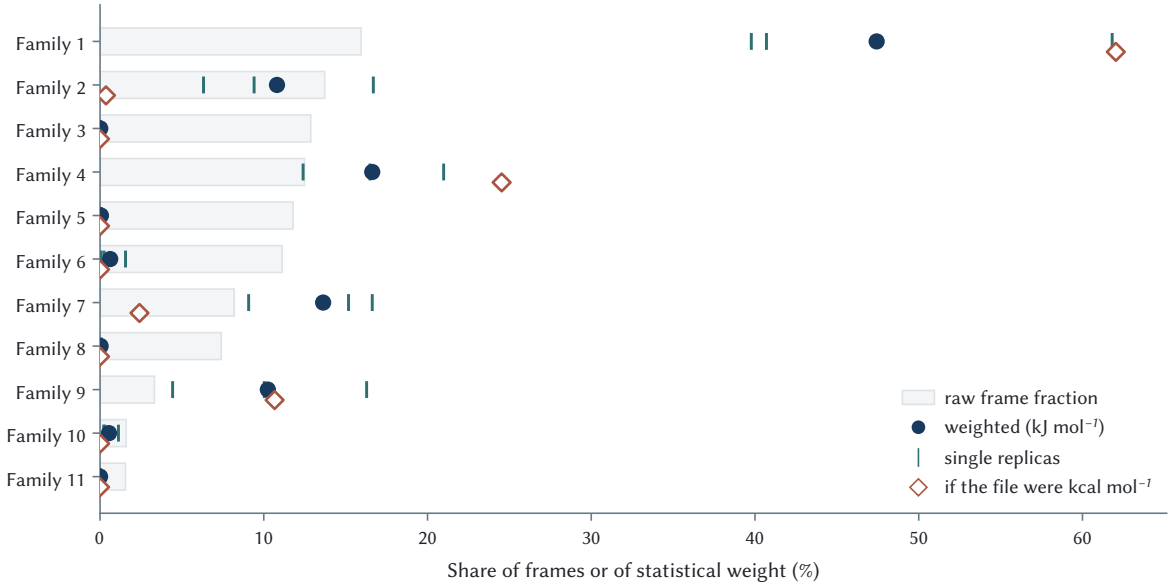


Figure 7 Family populations before and after reweighting, assuming the FES is in kJ mol^{-1}. Grey bars are raw frame fractions; navy dots are weighted populations; teal ticks are the three single replicas; open diamonds show the populations if the same file were in kcal mol^{-1}.

Table 7 Raw frame fractions and weighted populations of the density-peak families (%), under the assumed kJ mol^{-1} unit and, for comparison, if the file were in kcal mol^{-1}.

Family	Raw	Weighted	Replica range	If kcal mol^{-1}
1	15.9	47.4	39.8–61.8	62.0
2	13.7	10.8	6.3–16.7	0.4
3	12.9	0.0	0.0–0.0	0.0
4	12.5	16.6	12.4–21.0	24.5
5	11.8	0.1	0.0–0.1	0.0
6	11.1	0.6	0.1–1.6	0.0
7	8.2	13.6	9.1–16.6	2.4
8	7.4	0.0	0.0–0.1	0.0
9	3.3	10.2	4.4–16.3	10.7
10	1.6	0.5	0.2–1.1	0.0
11	1.6	0.0	0.0–0.0	0.0

Replica range: lowest and highest of the three single-replica populations. A moving-block bootstrap was also run, but with blocks of 21,687–63,485 frames (twice the largest statistical inefficiency) each replica holds only two to five blocks and the resulting intervals were not centred on the estimates, so they are not reported.

6.6 What changes when SASA is removed

All 50 Gaussian-mixture fits converged; the run took 7.01 s with a peak of 660.2 MiB. Paired fits with and without SASA share observations and seed, so their ARI measures the feature effect, while fits within one feature space differ only in initialization (Figure 8; values in Table 13, Appendix C).

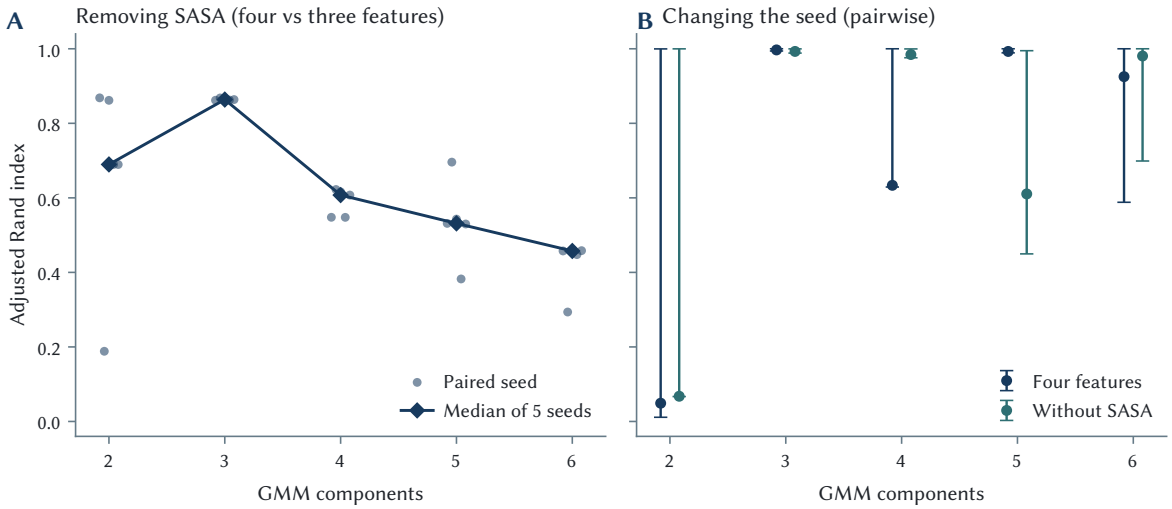


Figure 8 The independent GMM comparison. (A) ARI between four-feature and SASA-removed assignments for five paired seeds; diamonds are medians. (B) Within-feature-space ARI for ten seed pairs; markers are medians and bars span the observed minimum and maximum. Every model uses the same fitting subset, and evaluation includes fitting rows.

At three components the cross-feature ARIs are 0.862–0.868, and a median 95.3% of frames keep corresponding labels; within-space agreement is 0.994–1.000 with four descriptors and 0.988–1.000 without SASA. At two components, by contrast, within-space ARIs drop as low as 0.011 and 0.067, so a single fit would have been misleading (all seed pairs in Figure 14). Seed stability alone does not single out three components, since four-feature fits at five components (0.989–1.000) and three-feature fits at four (0.976–1.000) are equally reproducible, and the Bayesian information criterion (BIC) of each feature space keeps decreasing up to six. What distinguishes three components is that the partition is stable in both feature spaces and agrees across them; at the other counts the median cross-feature ARI falls to 0.457–0.690. At three components, removing SASA changes the labels of about 5% of frames, so SASA is largely but not entirely redundant for this model.

6.7 Interpretation

Correlation alone does not settle a feature choice: it summarizes the whole sample, whereas clustering assigns individual frames using all retained features, and the answer here depends on the model. The two clustering approaches also behave differently. The three-component Gaussian mixture is almost insensitive to its seed, whereas the density-peak partition changes substantially with the seed, the parameters and the amount of data used for discovery. The density-peak families should therefore be read as one reproducible outcome of a stated configuration rather than a unique decomposition of the ensemble, and their weighted populations depend in addition on the assumed FES unit. Whether these families correspond to those reported in the primary article has not yet been compared.

7 Skills acquired and critical reflection

This chapter compares the outcomes with the internship objectives and reflects on the skills acquired.

7.1 Objectives and outcomes

Table 8 compares the aims of the training project and the three parts of the final task with what was achieved, pointing to the evidence for each.

Table 8 Internship objectives compared with outcomes.

Objective	Status	Evidence
Develop computational and analytical skills for intrinsically disordered proteins (training project)	Achieved	Preparatory study and practice (Chapter 3 and Section 5.2); analyses of Chapter 6
Run the script and reproduce the published results	Achieved for the clustering run; populations conditional	Script audited and corrected; clustering run on all frames with the published parameters (Section 5.3); families not yet compared with those reported in the primary article; weighted populations computed under an assumed FES unit (Section 6.5)
Vary the parameters	Achieved	Seed, cutoff, silhouette, cluster-size and scaling variants on the real data (Sections 5.4 and 6.4)
Propose and explore an independent framework	Achieved	50-fit Gaussian-mixture feature experiment (Sections 5.5 and 6.6), with PCA and temporal diagnostics (Section 6.3)
Report of about 20 pages (minimum 15)	Achieved	This document: 21 pages of main text (Chapters 1–8), with bibliography and appendices

7.2 Skills acquired

Table 9 summarizes the skills exercised during the internship and where each is documented. It records activities rather than self-assessed levels; the assessment of the internship is made by the host supervisor.

7.3 Related personal projects

I am more at home with code than with molecular software, and during the internship I wanted to learn GROMACS and VMD quickly by working with AI assistants. Without a dedicated interface, such an assistant can only operate GROMACS or VMD through the screen. I therefore built two personal projects, with AI coding assistance: VMD MCP and GROMACS MCP. Both are Python servers that follow the Model Context Protocol (MCP), an open protocol through which software assistants call external tools with structured inputs and outputs, so that an assistant can call the two programs directly. The first versions date from early July 2026, during the simulation block. `vmd-mcp` exposes 12 tools for structure loading, measurement and headless rendering; `gromacs-mcp` exposes 31 tools for system preparation, simulation runs and analysis, and has a tagged v0.2.0 release

Table 9 Skills exercised during the internship and the evidence in this report.

Skill area	Activity and evidence
Protein structure and intrinsic disorder	Preparatory study; background of Chapter 3
Molecular simulation software	GROMACS and VMD practice (Section 5.2)
Enhanced sampling and reweighting	Reading WT-MetaD output; final-bias weights and their unit dependence (Section 3.4)
Scientific Python and data validation	Replica parsing, FES grid checks, frame identities (Chapter 4 and Section 5.3)
Time-series diagnostics	Autocorrelation and effective sample size (Sections 4.5 and 6.3)
Clustering and model comparison	Density-peak workflow; Gaussian mixtures, ARI, BIC (Sections 5.3, 6.4 and 6.6)
Software testing and reproducibility	38 tests, run manifests, continuous integration (Sections 5.3 and 5.6 and Appendix E)
Scientific writing	This report in LaTeX, with figures generated from saved tables

and continuous integration. They are separate from the Aβ40 analysis: no Aβ40 data in this report were produced with them. Source excerpts, screenshots and a map of the tools are in Appendix F (Figures 16 to 18).

7.4 Critical reflection

The hardest part of the internship came at the start, and it was practical rather than statistical. I am a data-analysis student and began with little background in biology and none in GROMACS or VMD: there were many commands and options, and at first I did not know which to use or in what order. The co-supervisor's explanations and an online GROMACS tutorial, which I worked through at home, got me past this preparatory phase, and the work became much easier afterwards. GROMACS itself did not take a large share of the 75 hours, and we did not go into its finer details.

Working with supplied files raised a second kind of difficulty: the script did not run as delivered, an intermediate version was missing, and the free-energy surface did not state its unit. Each was recorded rather than patched over: the script problems in Table 4, and the missing unit and replica combination as an explicit assumption (Section 6.5). I have not yet raised the unit question with the supervisors; it is the first of the next steps in Chapter 8.

On the statistical side, the idea I found hardest was the effective sample size. This was also my first work with biological data: I first had to learn what each descriptor measures, and the table was larger than the data sets of my coursework.

What went well was turning a single observation into a controlled experiment. The strong R_g–SASA correlation could have justified dropping a feature; testing that decision with fixed samples and repeated seeds showed that the answer depends on the number of components: with three, removing SASA changes few assignments (cross-feature ARI 0.862–0.868), whereas at the other counts the median ARI falls to 0.457–0.690 (Section 6.6). The practical lesson is to ask early for the provenance of every input file: one missing unit made the weighted populations conditional.

Some things remain unfinished: the families have not been compared with those reported in the primary article, and the FES unit is still an assumption (Table 5). Looking back, I would go deeper into the biology and spend more time exploring structures in VMD. I chose this internship to strengthen my background in biology, because I would like to move towards bioinformatics, and I am grateful to the supervisors for accepting me.

8 Conclusions and future work

This chapter summarizes what was achieved and what the next steps should be.

The internship turned a supplied clustering script and data set into an inspectable, tested analysis. The audit identified problems in parsing, residual-cluster handling, FES interpolation, replica pooling and weighted summaries, and the modular reimplementation corrects them with 38 passing tests while preserving the original sources. All three parts of the final task were carried out; Table 8 records the status of each.

The corrected workflow runs the published density-peak clustering on all 300,003 frames with the published parameters. It finds 11 families that differ mainly in secondary structure and are shared unevenly between replicas, and the parameter study shows that the partition depends strongly on the seed, the parameters and the data used for discovery. Under an assumed $\mathrm{kJ\,mol^{-1}}$ FES unit, reweighting concentrates 77.7% of the equilibrium weight on three families and almost removes the helix-rich ones, a pattern that would also hold if the file were in $\mathrm{kcal\,mol^{-1}}$. The independent Gaussian-mixture experiment shows that the strong R_g–SASA correlation does not settle the feature choice: removing SASA largely preserves the three-component partition (ARI 0.862–0.868) but not the others (median ARI 0.457–0.690).

The next steps, in order of priority, are:

1. confirm with the authors the FES unit and how the replicas were combined; the populations of Section 6.5 can then be recomputed immediately from the saved labels;
2. repeat the seed study with the 20-seed profile that the full configuration already defines, and compare the families with those reported in the primary article;
3. repeat the feature experiment on disjoint time blocks, to measure the sampling sensitivity that the fixed fitting sample deliberately excludes [8];
4. contrast final-bias weights with time-dependent `metad.rbias` weights once the metadynamics settings are known [12, 14], and compare with DPA [28] and with an experimentally constrained Aβ40 ensemble [3];
5. with atomic trajectories and a frame map, extract representative structures and test coordinate-based methods such as ShapeGMM [29, 30].

Signatures

Curricular internship, 25 May–30 September 2026, 75 hours, Laboratory of Experimental and Computational Physics of Biosystems, University of Messina.

STUDENT

Ali Erkan Ocaklı

Signature

Place and date

HOST SUPERVISOR

Prof. Valeria Conti Nibali

Signature

Place and date

ACADEMIC TUTOR

Prof. David Barilla

Signature

Place and date

Bibliography

[1] R. van der Lee, M. Buljan, B. Lang, et al. "Classification of Intrinsically Disordered Regions and Proteins". In: *Chemical Reviews* 114.13 (2014), pp. 6589–6631. DOI: 10.1021/cr400525m.

[2] I. Cadenelli, A. Ciccolo, A. Tagliabue, G. Rossi, V. Conti Nibali, and D. Bochicchio. "The Monomeric Conformational Ensembles of Aβ40 and Aβ42 Encode Their Differential Amyloid Aggregation Propensity". In: *The Journal of Physical Chemistry B* 130.24 (2026), pp. 6035–6046. DOI: 10.1021/acs.jpcb.6c01000.

[3] K. Borthakur, T. R. Sisk, F. P. Panei, M. Bonomi, and P. Robustelli. "Determining accurate conformational ensembles of intrinsically disordered proteins at atomic resolution". In: *Nature Communications* 16.1, 9036 (2025). DOI: 10.1038/s41467-025-64098-3.

[4] M. J. Abraham, T. Murtola, R. Schulz, et al. "GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers". In: *SoftwareX* 1-2 (2015), pp. 19–25. DOI: 10.1016/j.softx.2015.06.001.

[5] J. Huang, S. Rauscher, G. Nawrocki, et al. "CHARMM36m: an improved force field for folded and intrinsically disordered proteins". In: *Nature Methods* 14.1 (2017), pp. 71–73. DOI: 10.1038/nmeth.4067.

[6] G. A. Tribello, M. Bonomi, D. Branduardi, C. Camilloni, and G. Bussi. "PLUMED 2: New feathers for an old bird". In: *Computer Physics Communications* 185.2 (2014), pp. 604–613. DOI: 10.1016/j.cpc.2013.09.018.

[7] J. D. Chodera. "A Simple Method for Automated Equilibration Detection in Molecular Simulations". In: *Journal of Chemical Theory and Computation* 12.4 (2016), pp. 1799–1805. DOI: 10.1021/acs.jctc.5b00784.

[8] A. Grossfield, P. N. Patrone, D. R. Roe, A. J. Schultz, D. Siderius, and D. M. Zuckerman. "Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations [Article v1.0]". In: *Living Journal of Computational Molecular Science* 1.1, 5067 (2019). DOI: 10.33011/livecoms.1.1.5067.

[9] A. Barducci, G. Bussi, and M. Parrinello. "Well-Tempered Metadynamics: A Smoothly Converging and Tunable Free-Energy Method". In: *Physical Review Letters* 100.2, 020603 (2008). DOI: 10.1103/PhysRevLett.100.020603.

[10] PLUMED developers. *sum_hills*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/sum_hills.html (visited on 2026-11-12).

[11] PLUMED developers. *METAD: metad.bias, metad.rbias and c(t)*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/_m_e_t_a_d.html (visited on 2026-11-12).

[12] P. Tiwary and M. Parrinello. "A Time-Independent Free Energy Estimator for Metadynamics". In: *The Journal of Physical Chemistry B* 119.3 (2015), pp. 736–742. DOI: 10.1021/jp504920s.

[13] PLUMED developers. *REWEIGHT_METAD*. URL: https://www.plumed.org/doc-v2.9/user-doc/html/_r_e_w_e_i_g_h_t__m_e_t_a_d.html (visited on 2026-11-12).

[14] M. Bonomi, A. Barducci, and M. Parrinello. "Reconstructing the equilibrium Boltzmann distribution from well-tempered metadynamics". In: *Journal of Computational Chemistry* 30.11 (2009), pp. 1615–1621. DOI: 10.1002/jcc.21305.

[15] W. Kabsch and C. Sander. "Dictionary of protein secondary structure: Pattern recognition of hydrogen-bonded and geometrical features". In: *Biopolymers* 22.12 (1983), pp. 2577–2637. DOI: 10.1002/bip.360221211.

[16] A. Rodriguez and A. Laio. "Clustering by fast search and find of density peaks". In: *Science* 344.6191 (2014), pp. 1492–1496. DOI: 10.1126/science.1242072.

[17] P. J. Rousseeuw. “Silhouettes: A graphical aid to the interpretation and validation of cluster analysis”. In: *Journal of Computational and Applied Mathematics* 20 (1987), pp. 53–65. DOI: 10.1016/0377-0427(87)90125-7.

[18] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. “Array programming with NumPy”. In: *Nature* 585.7825 (2020), pp. 357–362. DOI: 10.1038/s41586-020-2649-2.

[19] P. Virtanen, R. Gommers, T. E. Oliphant, et al. “SciPy 1.0: fundamental algorithms for scientific computing in Python”. In: *Nature Methods* 17.3 (2020), pp. 261–272. DOI: 10.1038/s41592-019-0686-2.

[20] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. “Scikit-learn: Machine Learning in Python”. In: *Journal of Machine Learning Research* 12.85 (2011), pp. 2825–2830. URL: https://www.jmlr.org/papers/v12/pedregosa11a.html.

[21] Snakemake developers. *Snakefiles and Rules*. URL: https://snakemake.readthedocs.io/en/stable/snakefiles/rules.html (visited on 2026-11-12).

[22] NumPy developers. *Parallel random number generation*. URL: https://numpy.org/doc/stable/reference/random/parallel.html (visited on 2026-11-12).

[23] SciPy developers. *wasserstein_distance*. URL: https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wasserstein_distance.html (visited on 2026-11-12).

[24] SciPy developers. *RegularGridInterpolator*. URL: https://docs.scipy.org/doc/scipy/reference/generated/scipy.interpolate.RegularGridInterpolator.html (visited on 2026-11-12).

[25] C. J. Geyer. “Practical Markov Chain Monte Carlo”. In: *Statistical Science* 7.4 (1992), pp. 473–483. DOI: 10.1214/ss/1177011137.

[26] L. Hubert and P. Arabie. “Comparing partitions”. In: *Journal of Classification* 2.1 (1985), pp. 193–218. DOI: 10.1007/BF01908075.

[27] L. Wratten, A. Wilm, and J. Göke. “Reproducible, scalable, and shareable analysis pipelines with bioinformatics workflow managers”. In: *Nature Methods* 18.10 (2021), pp. 1161–1168. DOI: 10.1038/s41592-021-01254-9.

[28] M. d’Errico, E. Facco, A. Laio, and A. Rodriguez. “Automatic topography of high-dimensional data sets by non-parametric density peak clustering”. In: *Information Sciences* 560 (2021), pp. 476–492. DOI: 10.1016/j.ins.2021.01.010.

[29] S. Sasmal, T. Pal, G. M. Hocky, and M. McCullagh. “Quantifying Unbiased Conformational Ensembles from Biased Simulations Using ShapeGMM”. In: *Journal of Chemical Theory and Computation* 20.9 (2024), pp. 3492–3502. DOI: 10.1021/acs.jctc.4c00223.

[30] N. Michaud-Agrawal, E. J. Denning, T. B. Woolf, and O. Beckstein. “MDAnalysis: A toolkit for the analysis of molecular dynamics simulations”. In: *Journal of Computational Chemistry* 32.10 (2011), pp. 2319–2327. DOI: 10.1002/jcc.21787.

[31] CINECA. *Leonardo: HPC documentation*. URL: https://docs.hpc.cineca.it/hpc/leonardo.html (visited on 2026-11-12).

A Parameters and settings

Iterative density-peak discovery

Within each random group of about 30,000 frames, discovery proceeds as follows.

1. **Generate candidates.** Start discovery at $d_c = 1.3$, rank candidate centres by $\rho_i \delta_i$, and try 10 down to 2 centres.
2. **Check each cluster.** Require at least 300 members and a mean silhouette of at least 0.3 for acceptance.
3. **Choose the candidate.** Keep the one with the most acceptable clusters. A tie retains the earlier, higher-centre-count candidate.
4. **Continue on the remainder.** Remove accepted members and multiply d_c by 0.9 before the next iteration.
5. **Stop discovery.** Stop below 300 remaining points or after five consecutive unsuccessful iterations. Residual frames receive their final labels only in the final-assignment step (step 5 of Figure 3).

Table 10 Settings of the density-peak workflow and its weighted extension, with the execution profiles used. Full-profile values describe the runs reported in Sections 6.4 and 6.5.

Parameter	Value	Evidence / role
Replicas	3	Primary article; observed table segmentation
Analysis interval	1–3 µs	Published baseline; inclusive source-labelled ps boundaries
Temperature; bias factor γ	310 K; 10	Published weighting parameters; no new MD run
Molar gas constant	0.008314462618 kJ mol^{-1} K^{-1}	Implementation unit convention
CV conversions	$\alpha = n_\alpha/1.1$, $\beta = n_\beta/2.5$	Published descriptor scaling
Baseline features	n_α, n_β, R_g, SASA	Pooled min–max normalization
Discovery	$d_c = 1.3$, multiplied by 0.9 per iteration; 10 down to 2 centres; at least 300 frames; mean silhouette at least 0.3; stop after 5 failures	Published baseline (steps 1–5 above)
Distance storage	float32	Source-informed; blockwise construction
Consensus	Average linkage	Euclidean norm of the four marginal Wasserstein-1 distances
Elbow heuristic	$s = 5$; first 90% of merges	Supplied source heuristic
Final assignment	One nearest-centroid pass	No iterative refitting
Master seed	42	Default project configuration; recorded per run
Profiles (smoke / medium / full)	Groups of 3,000 / 10,000 / about 30,000 frames; 2 / 3 / all groups	Smoke and medium are reduced profiles; full uses 10 groups covering all 300,003 frames
Seed study	2 smoke; 5 medium; 5 full	Initialization seeds only; seed 42 is the reference. The configured 20-seed full study was not run

Table 10 Settings of the density-peak workflow and its weighted extension (continued).

Parameter	Value	Evidence / role
Parameter study	$d_c \times 0.9$ and $\times 1.1$; silhouette 0.25 and 0.35; minimum size 200 and 400	One change at a time; medium and full profiles
FES energy unit	kJ mol^{-1} (assumed)	Not confirmed by the authors; set in the configuration file `paper_reproduction_full_assumed.yaml`
Bootstrap	500 draws; blocks of twice the largest statistical inefficiency	Run but not reported: two to five blocks per replica gave intervals not centred on the estimates (Table 7)
Resource policy	Bounded sequential local workers	Full-profile limits 24 h and 24 GiB; measured for the baseline run: 14 min, 3.8 GiB

Settings of the Gaussian-mixture experiment

Table 11 lists the settings of the independent GMM experiment, which are separate from the published density-peak baseline; the protocol is described in Section 5.5.

Table 11 Settings of the completed GMM experiment, as recorded in its run manifest.

Parameter	Value	Evidence / role
Fit sample	1,000 per replica	3,000 rows; shared across models
Sampling seed	20261007	Local NumPy generator
Feature spaces	Four descriptors; omit SASA	Same scaling for shared features
Scaling	Fit-sample mean and SD	No evaluation-data refit
Components	2, 3, 4, 5, 6	All counts retained in comparison
Initialization seeds	42, 43, 44, 45, 46	Paired across representations
Initializations per fit	1	50 total fitted models
Covariance	Full	Regularization 10^{-6}
Convergence settings	500 iterations; tolerance 10^{-3}	All fits converged numerically
Evaluation	300,003 rows	Includes fit rows; descriptive comparison
Comparisons	5 paired-seed ARIs between feature spaces; 10 seed pairs within each	All 300,003 frames, fitting rows included
Resource limits	600 s; 4 GiB	Sequential local worker

B Mathematical derivations and estimator checks

Gaussian hills and bias construction

The accumulated potential can be written as a sum of local Gaussian hills:

$$V(s,t) = \sum_{t_k<t} h_k \exp\left[-\frac{1}{2}\sum_a \frac{(s_a - s_a(t_k))^2}{\sigma_a^2}\right]. \tag{7}$$

In Equation (7), s is the CV vector, t_k a deposition time, h_k the hill height and σ_a its width along coordinate a. Well-tempered heights follow $h_k = h_0 \exp[-V(s(t_k), t_k)/(k_B \Delta T)]$ in molecular energy units, with $\Delta T = (\gamma - 1)T$; molar units require R instead of k_B.

Sign of the final-bias correction

Under a static bias V, the observed configuration density is proportional to $\exp[-\beta_T(U + V)]$. Multiplying by $\exp(\beta_T V)$ recovers relative canonical density. The WT relation $V \simeq -(1 - 1/\gamma)F + C$ therefore gives the negative free-energy sign in Equation (2). Applying $\exp(-\beta_T V)$ would reverse the correction. This derivation assumes that using a final bias is appropriate for the analysed interval; it is not a proof that the supplied trajectories satisfy that assumption.

Kish effective sample size

Unequal statistical weights require a separate diagnostic:

$$N_{\mathrm{eff,Kish}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{8}$$

Equation (8) measures weight concentration without correcting temporal dependence, so it is not combined with the correlation ESS of Table 12. Under the assumed FES unit of Section 6.5 it ranges from 3,277 to 9,551 per replica of 100,001 frames.

Equal-replica mixture and cluster moments

Let $\widetilde{w}_{ri}$ sum to one within replica r. The pooled expectation of A is $(1/3)\sum_r \sum_i \widetilde{w}_{ri} A_{ri}$. Cluster relevance uses A equal to a membership indicator. For a conditional cluster mean, divide the pooled weighted sum of the descriptor by the pooled membership mass. Computing three conditional means and averaging them equally instead would answer a different question when cluster populations differ across replicas.

Bilinear interpolation test

For a cell with fractional coordinates $u, v \in [0, 1]$, interpolation combines its four corner values with weights $(1-u)(1-v)$, $u(1-v)$, $(1-u)v$, and uv. Their sum is one. Any surface $a + b\alpha + c\beta + d\alpha\beta$ is reproduced exactly within floating-point tolerance. The test suite includes such a mixed-term surface, reordered input rows, boundaries and invalid-grid cases, providing stronger evidence than checking a single grid node.

C Supplementary diagnostics

This appendix collects the figures and tables behind Sections 6.3 and 6.6. Every descriptor diagnostic uses the unweighted, biased sample of 300,003 frames (three replicas, 1–3 µs); weights enter only in Section 6.5. The saved tables behind these figures belong to run 20260921T011959-583b4421.

Descriptor distributions and relationships

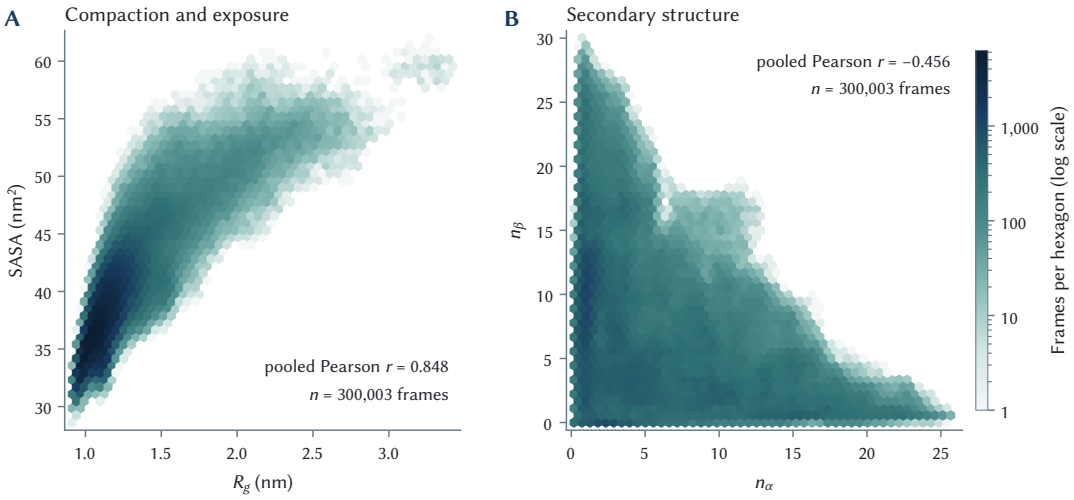


Figure 9 Unweighted descriptor landscape of the pooled sample (300,003 frames, three replicas, 1–3 µs). (A) Radius of gyration against SASA. (B) n_α against n_β. Shading is the log number of frames per hexagonal cell on a shared scale.

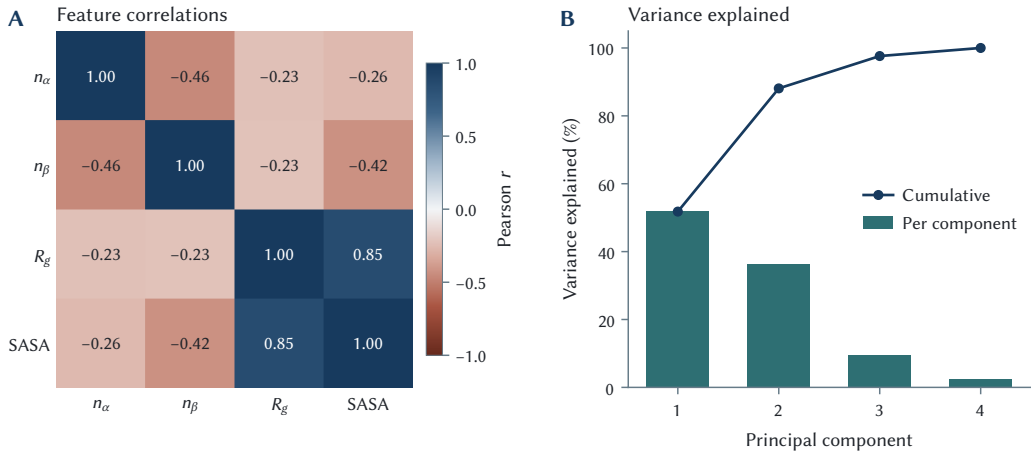


Figure 10 Relationships between the four descriptors in the pooled, unweighted sample. (A) Pearson correlation matrix. (B) Variance explained by standardized PCA; the line shows cumulative variance.

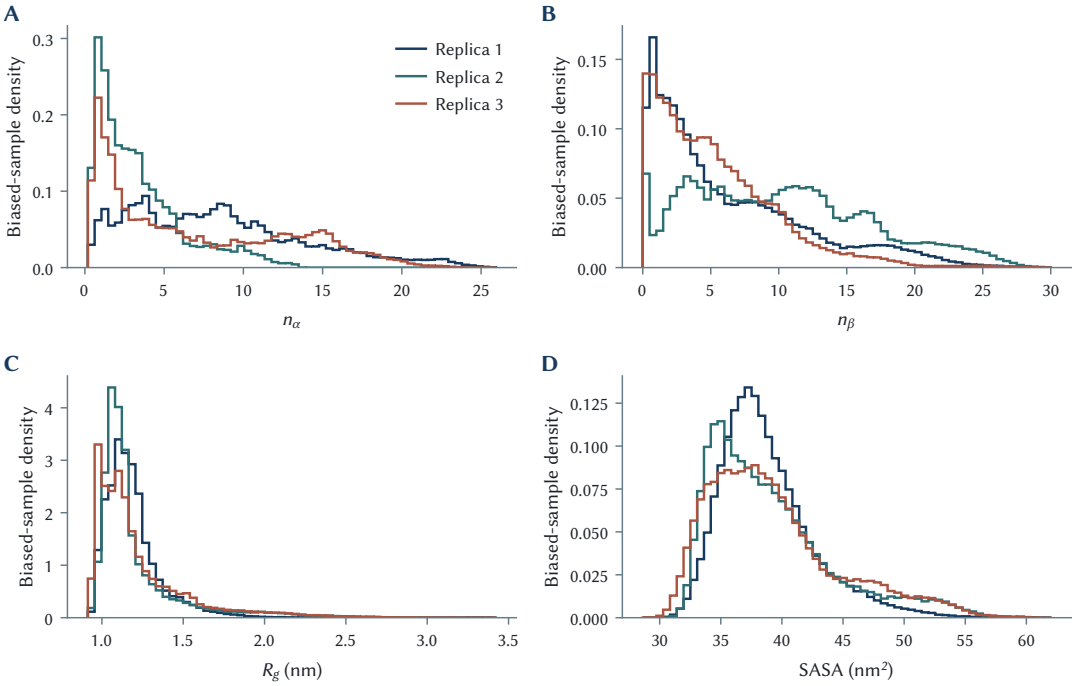


Figure 11 Per-replica distributions of the recorded descriptors. Each descriptor uses common bin edges, with separate density normalization for each replica. The curves retain the simulation bias and are not equilibrium distributions. Units follow the source labels. Panels: (A) n_α, (B) n_β, (C) R_g, (D) SASA.

Dependence between saved frames

Table 12 Correlation-time estimates for all twelve replica–descriptor series. Every series is flagged as unreliable: $N_{\mathrm{eff,corr}}$ is below the required 50 for all of them, and the half-window shift exceeds 0.25 standard deviations in 11 of 12. g is the statistical inefficiency; the half-window shift is the absolute difference of the half-series means divided by the full-series standard deviation. Time is biased trajectory time.

Replica	Descriptor	g	τ_{int} (ns)	$N_{\mathrm{eff,corr}}$	Half shift / SD
1	n_α	22,067	220.7	4.5	1.28
1	n_β	11,079	110.8	9.0	0.83
1	R_g	3,679	36.8	27.2	0.27
1	SASA	3,490	34.9	28.7	0.36
2	n_α	6,259	62.6	16.0	0.37
2	n_β	10,843	108.4	9.2	0.41
2	R_g	3,951	39.5	25.3	0.33
2	SASA	5,325	53.2	18.8	0.17
3	n_α	31,742	317.4	3.2	1.61
3	n_β	5,880	58.8	17.0	0.71
3	R_g	6,259	62.6	16.0	0.88
3	SASA	9,958	99.6	10.0	0.87

The half-window mean-shift diagnostic is descriptive: a large shift can arise from slow fluctuations, nonstationarity or changing occupation of sampled regions, and a series is flagged when the shift exceeds 0.25 standard deviations, an explicit project heuristic. A small shift alone would not establish convergence, and no threshold outcome can recover an unbiased time scale from WT-MetaD time.

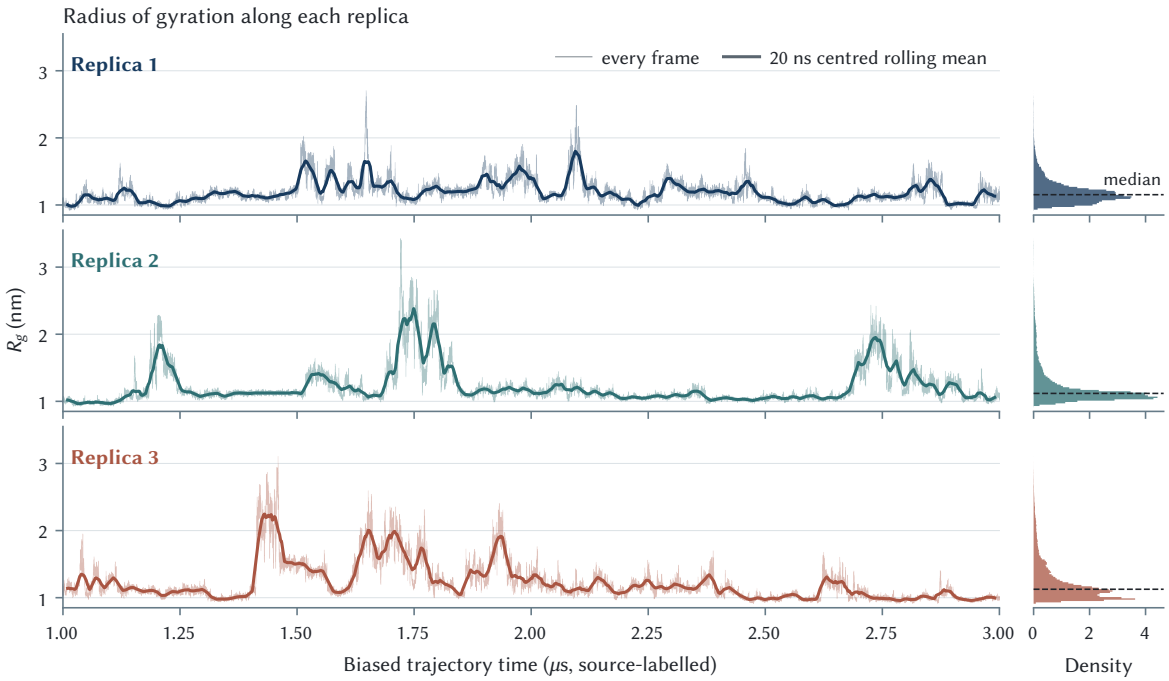


Figure 12 Radius of gyration along the biased trajectory of each replica. Thin lines show every frame and bold lines a centred 20 ns (1,000-frame) rolling mean. The right-hand histograms are unweighted densities with dashed lines at the medians (1.153, 1.118 and 1.126 nm).

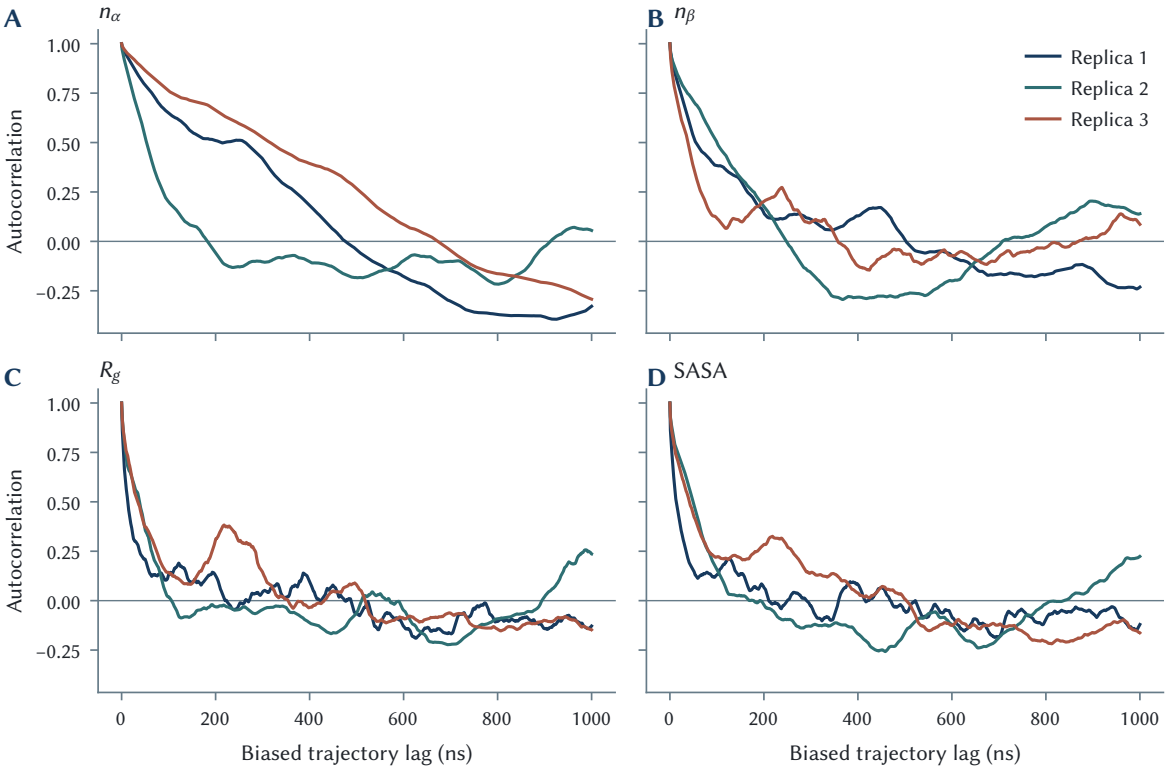


Figure 13 Autocorrelation functions calculated separately for each replica and descriptor. For readability the figure displays a subset of lag points over the first half of the available window; estimation uses the full ACF sequence. The horizontal axis is biased trajectory time. Panels: (A) n_α, (B) n_β, (C) R_g, (D) SASA.

Gaussian-mixture comparison

Table 13 lists the values plotted in Figure 8, and Figure 14 shows every pairwise seed ARI behind the within-feature-space ranges reported there.

Table 13 Agreement between the four-feature and SASA-removed Gaussian-mixture partitions over the five paired initializations (values plotted in Figure 8).

Components	Median ARI, four vs three features	Observed range
2	0.690	0.188–0.868
3	0.864	0.862–0.868
4	0.607	0.548–0.622
5	0.531	0.382–0.696
6	0.457	0.294–0.461

Component counts were specified for comparison, not selected as physical numbers of states.

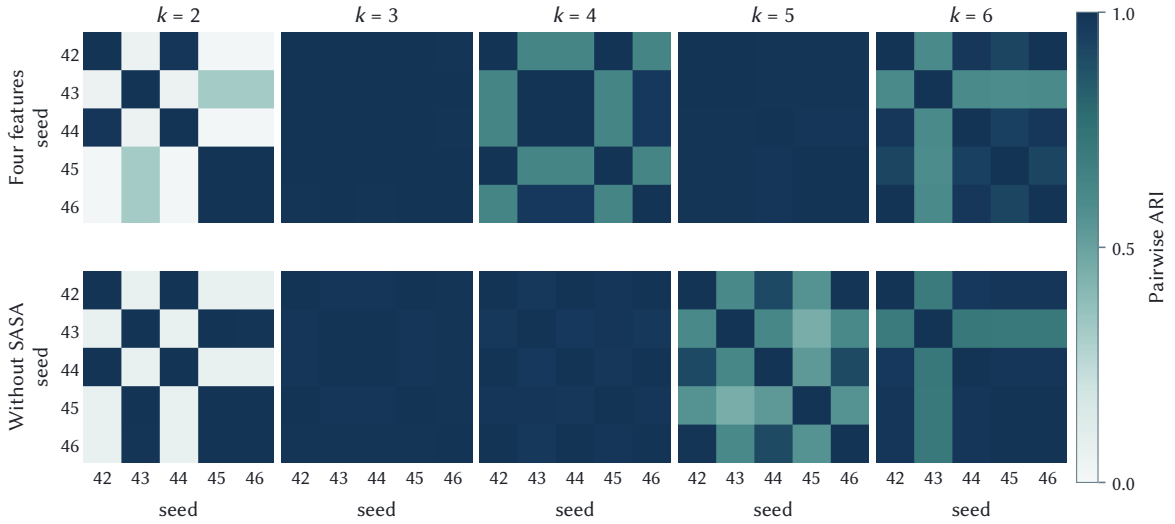


Figure 14 All pairwise seed ARIs from the GMM experiment. Columns show each tested component count; rows distinguish the four-feature and SASA-removed representations. The diagonal is identity. All fits converged, so no entries are omitted; convergence of the optimizer does not guarantee the same partition.

D Code audit summary

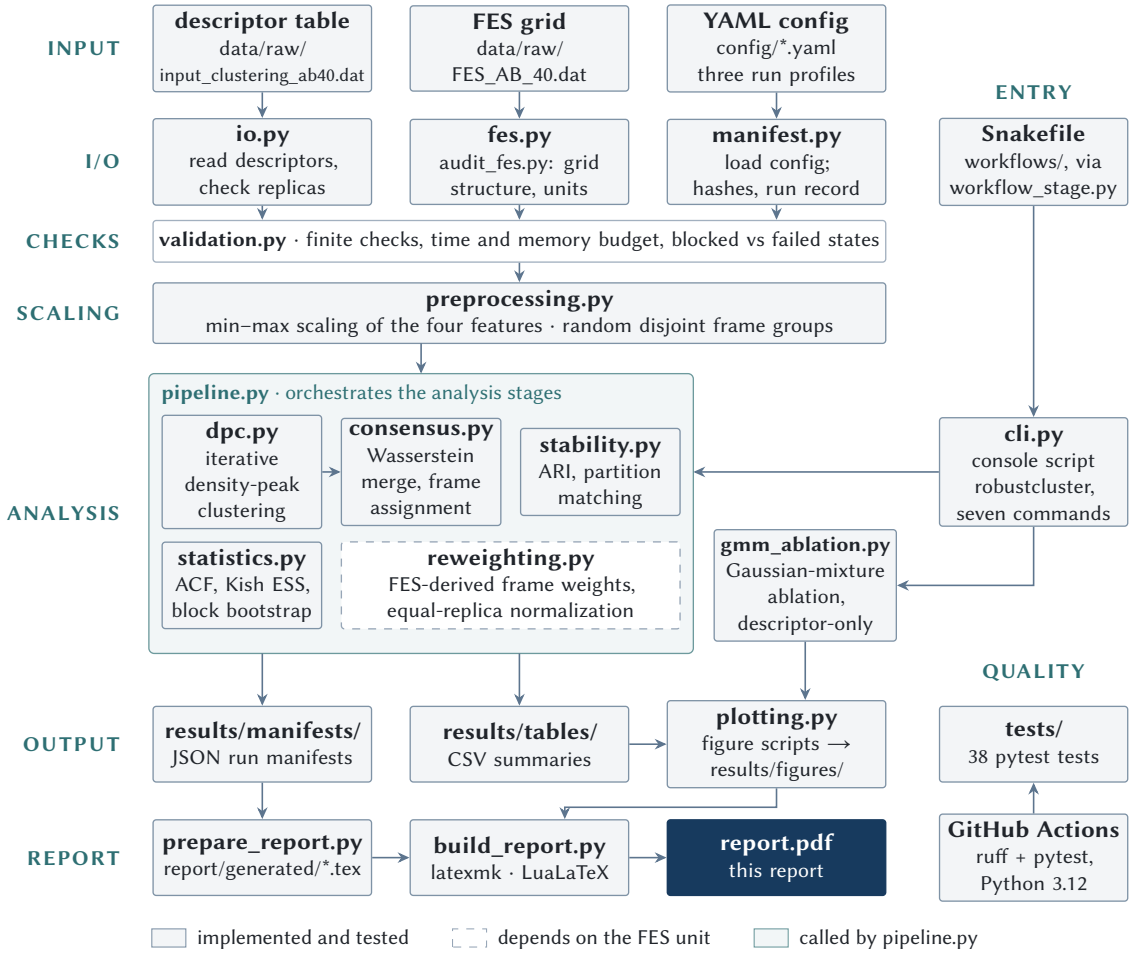


Figure 15 Architecture of the `robustcluster-abeta` package. Inputs pass through validation and scaling to the analysis modules orchestrated by `pipeline.py`; results go to manifest-backed tables and figures and then into this report. `gmm_ablation.py` is the separate descriptor-only experiment of Section 5.5. The dashed module depends on the FES unit and was run under the assumption of Section 6.5.

Two numerical tests

Two small examples from the 38-test suite show how a check can identify a numerical error without relying on the output of the full pipeline.

Interpolation with a known answer. The derivation is in Appendix B. An analytic bilinear surface includes a mixed-coordinate term, so a query between nodes tests both axes together. Reordering its rows must preserve the interpolated answer. Deleting or duplicating a grid point, or querying outside the bounds, must instead raise an error. The same fixture therefore checks both a valid calculation and the rejection of inputs for which that calculation would be inappropriate.

Normalization with an invariant. An arbitrary additive FES constant must disappear after weight normalization. A second check gives two replicas unequal numbers of frames and requires

each to contribute half of the pooled mass. This catches pooling by row count, which would look correct on three equal-length replicas. Other weight tests check the formula's sign and weighted moments.

The remaining checks cover parsing, scaling, residual exclusion and final assignment. Renaming cluster labels must leave partition agreement unchanged; changing evaluation extremes must not alter a scaler fitted on fixed training rows. These checks isolate failures that could otherwise be mistaken for a change in the clustering method. The scaling function behind the last of them is shown below.

Fit once, reuse the same scale RobustCluster-Aβ / Python

```
def fit_scaler(fit_values, values):
    fit_values, values = finite(fit_values), finite(values)
    mean, scale = fit_values.mean(axis=0), fit_values.std(axis=0)
    if np.any(scale <= 0):
        raise InvalidInput("Constant fit feature")
    return (values - mean) / scale, {"mean": mean.tolist(), "scale": scale.tolist()}
```

A synthetic end-to-end fixture exercises discovery, consensus, weighting, summaries and figure generation. Synthetic GMM cases check repeatability and missing-family handling. An autoregressive fixture probes the correlation estimator; an archive test prevents an unrelated enclosing Git repository from supplying false provenance. The fixtures establish software behaviour on known inputs (Scope 5).

The audit is documented in four project files: the inventory (immutable source hashes, table dimensions and missing inputs), the method map (each step linked to paper evidence, source behaviour, implementation and tests), the decision log (scientific category and expected effect of each change) and the open-questions register, which keeps uncertain provenance from becoming a hidden default.

An intermediate version of the supplied script, mentioned in the materials, could not be located; the audit therefore compares the supplied original directly with the reimplementation and reconstructs no intermediate version. The original is excluded from execution and from lint collection because its syntax error and top-level writes are part of the preserved evidence; the new code is tested independently. The raw table and the original script are not redistributed until their owners give permission.

E Reproducibility and software environment

The project source, tests, configurations and this report are available at RobustCluster-Aβ on GitHub. The Python package and LaTeX sources can be inspected separately or downloaded together using GitHub's *Code / Download ZIP* option. The repository README and `REPRODUCIBILITY.md` distinguish code verification and report regeneration from analyses that need the restricted inputs.

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements-lock.txt
python -m pip install --no-deps -e .
pytest -q
ruff check src tests scripts
robustcluster validate --config config/paper_reproduction.yaml
robustcluster diagnostics --config config/robustness.yaml
robustcluster cluster --config config/paper_reproduction_full.yaml --profile full
robustcluster reproduce --config config/paper_reproduction_full_assumed.yaml --profile full
```

The raw file must first be obtained and checked against its documented SHA256. The standalone commands and Snakemake workflow use the same package functions. The seed and parameter studies add `--experiment seed-pilot` or `--experiment parameters` to the `cluster` command with `config/robustness_full.yaml`, and the GMM experiment uses the `gmm-ablation` command with its own configuration. The report imports successful diagnostics, FES-audit, GMM, DPC-clustering, weighted-reproduction and figure manifests with matching hashes. The generated editorial artwork and the browser screenshots have separate provenance records. LuaLaTeX, Biber and latexmk build the report.

The full-profile configurations record the resource estimate that justified enabling them (at most 2 h and under 10 GiB, extrapolated from a medium-profile run of 36 s and 0.70 GiB). The full baseline run then took 14 min with a peak of 3.8 GiB. The configuration with the assumed FES unit states that assumption in its header and in every manifest it produces; once the unit is confirmed, only its two FES fields need to change. Outputs that would redistribute the FES or per-frame descriptor values stay local, with their hashes kept in the manifests. No calculation in this report used a high-performance computing (HPC) system such as Leonardo [31].

Software environment

Table 14 lists the software versions recorded in the manifest of the descriptor diagnostics. The third-party versions are identical in the GMM, DPC-clustering and weighted-population manifests; the package version is 0.1.0 in the diagnostics and GMM manifests and 0.3.0 in the two later ones. The versions do not describe the original authors' simulation environment.

Table 14 Software versions recorded in the manifest of the descriptor diagnostics.

Component	Version in diagnostic manifest
Python	3.12.6
numpy	2.5.3
scipy	1.18.1
scikit-learn	1.9.1
pandas	2.3.3
matplotlib	3.11.2
robustcluster-abeta	0.1.0

F Personal projects and source-code evidence

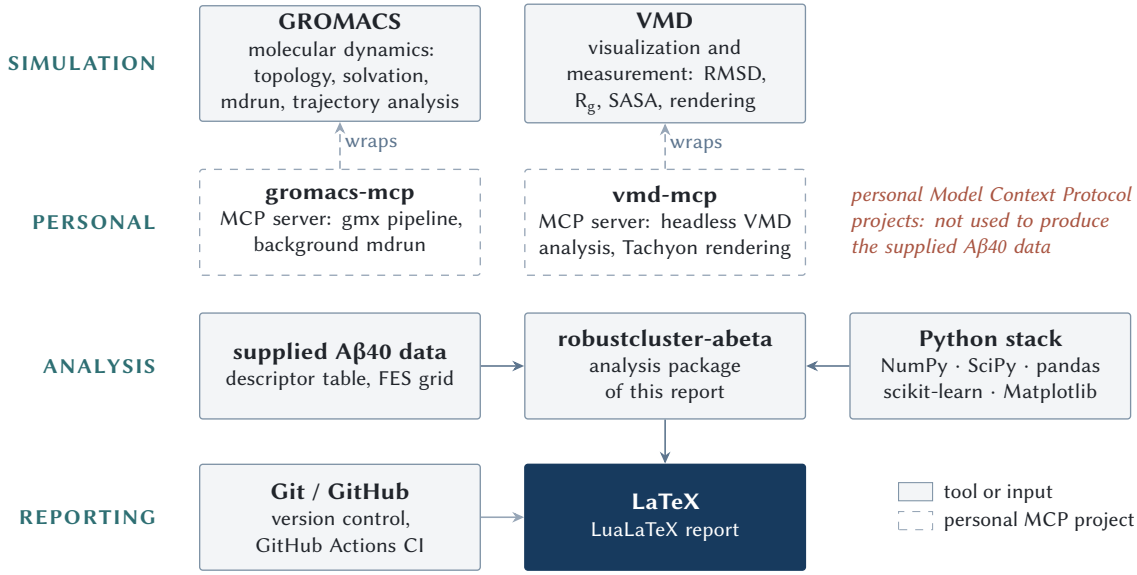


Figure 16 Tools used during the internship. GROMACS and VMD belong to the simulation block; the Python stack, Git/GitHub and LaTeX produced the analysis and this report. Dashed boxes are personal Model Context Protocol servers, which were not used to produce the supplied Aβ40 data.

The images below are unaltered browser captures of public, commit-specific GitHub source pages taken on 7 October 2026. They document the implementation, not any simulation run. Each is preceded by a transcription of the same lines as searchable text. The VMD wrapper passes an atom selection and a mass-weighting option to VMD and returns frame-indexed values; the GROMACS wrapper checks its paths and records output in a log.

Measure each saved frame — VMD MCP / source lines 250–266

```
250      weight = "weight mass" if mass_weighted else ""
251      tcl = f"""
252  {_load_block(structure, trajectory)}
253  set sel [atomselect $mol "{_tcl_quote(selection)}"]
254  set n [molinfo $mol get numframes]
255  for {{set i 0}} {{$i < $n}} {{incr i}} {{
256    $sel frame $i
257    puts "{MARKER} frame=$i rgyr=[measure rgyr $sel {weight}]"
258  }}
259  quit
260  """
261      res = _run_tcl(tcl)
262      series = [
263          {"frame": int(m["frame"]), "rgyr": float(m["rgyr"])}
264          for m in res.get("markers", [])
265          if "rgyr" in m
266      ]
```

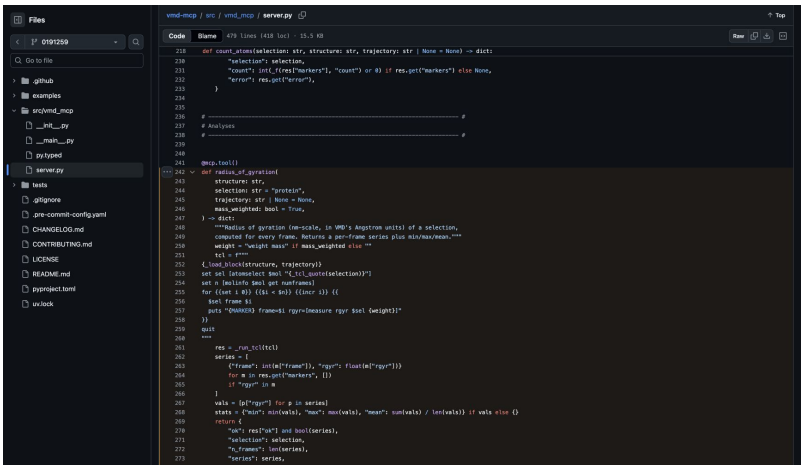


Figure 17 VMD MCP source at commit 0191259: the gyration wrapper constructs a Tcl frame loop and parses marked values. The screenshot shows the repository tree, source path and real line numbers. VMD reports lengths in ångström, so its values must be converted before comparison with the nanometre values of this report. The inspected source is available online.

Validate before starting a job GROMACS MCP / source lines 1452–1467

```
1452        wd, err = _workdir_or_error(workdir)
1453        if err:
1454            return err
1455        assert wd is not None
1456        if ntomp is not None and ntomp <= 0:
1457            return _error("ntomp must be positive when provided")
1458        if nsteps is not None and nsteps < 0:
1459            return _error("nsteps must be non-negative when provided")
1460        safe_tpr = _safe_file_arg(wd, tpr, "tpr", must_exist=True)
1461        safe_deffnm = _safe_deffnm(deffnm)
1462        args = [GMX_BIN, "mdrun", "-v", "-s", safe_tpr, "-deffnm", safe_deffnm]
1463        if ntomp is not None:
1464            args += ["-ntomp", str(ntomp)]
1465        if nsteps is not None:
1466            args += ["-nsteps", str(nsteps)]
1467        args += _validate_arg_tokens(extra_args, "extra_args")
```

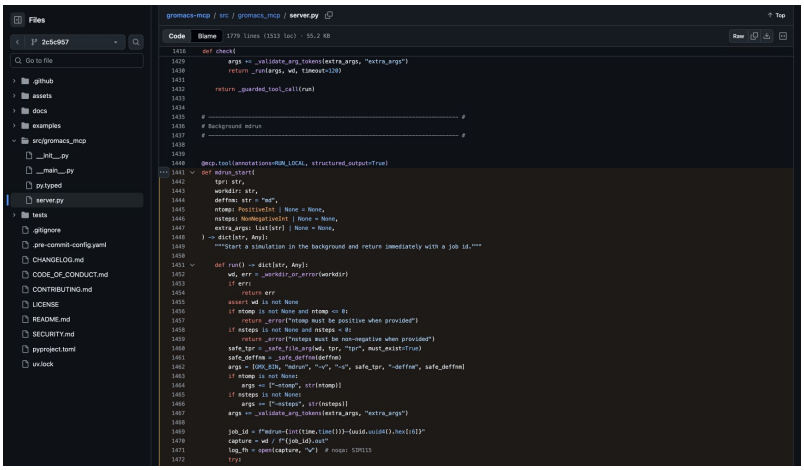


Figure 18 GROMACS MCP source at commit 2c5c957: the background-run wrapper validates inputs, constructs the `mdrun` command and prepares an output log. This source view does not establish that an Aβ40 simulation was launched through the tool. The inspected source is available online.